

An ontology-based framework for automatic building energy modeling with thermal zoning

Zhaoji Wu^a, Jack C.P. Cheng^{a,*}, Zhe Wang^{a,*}, Helen H.L. Kwok^b

^a Department of Civil and Environmental Engineering, The Hong Kong University of Science and Technology, Hong Kong Special Administrative Region

^b Institute for the Environment, The Hong Kong University of Science and Technology, Hong Kong Special Administrative Region

ARTICLE INFO

Keywords:

Building energy modeling
Ontology
Data integration
Thermal zone
Model translation

ABSTRACT

Building Energy Modeling (BEM) has wide application in building design and operation stages. Developing BEM requires numerous data from multiple sources, and therefore is time consuming and expertise demanding. Building Information Modeling can provide information needed for BEM, but a seamless transfer of BIM to BEM is not available yet. This study proposes an ontology-based automatic framework aimed at integrating multiple data sources and automatically generating BEM models, in which the ontology model could integrate data from weather, building, internal heat gain, and Heating, Ventilation, and Air Conditioning system. We developed a cross-domain and rule-based reasoning method for thermal zoning, and an automatic ontology model to BEM model translation method using instance-based mapping and dynamic data conversion. We demonstrated our approach by developing a BEM model for one floor of a campus building. The results showed that the proposed framework could automatically generate the model that has the potential to reduce the modeling time by over 99% comparing with manual modeling.

1. Introduction

The operation of buildings accounted for 30% of global final energy consumption and 27% of total energy sector emissions in 2021 [1]. More attention is being paid to building energy efficiency, especially under the regulatory trend for carbon neutrality. Building energy simulation is becoming a prerequisite of building energy management in many countries. Calibrated building energy simulation could assist building managers in predicting energy demands and adopting optimal control strategies. Further, with accurate simulation results, building managers can establish baselines and calculate retrofit savings.

Building energy simulation belongs to the forward (classical) approach (i.e., physical modeling approach) of Building Energy Modeling (BEM) [2] and relies on an accurate physical description of a building and thermodynamics rules. To achieve detailed energy modelling and analysis, the simulation model requires detailed input data and thermodynamic rules [3]. So far, the thermodynamic models have been supported by mature building energy simulation software, e. g., EnergyPlus [4] and eQuest [5]. However, the input data need to be still provided by users. Such input data include: location, weather data, building geometry, construction, materials, internal loads, Heating,

Ventilation, and Air Conditioning (HVAC) systems, operating schedules, etc. [6]. The lack of precise input data results in low accuracy of prediction [7], and the required data are of often different types and come from various information sources. For example, building geometries may be provided by Building Information Modeling (BIM) while the operation strategies and schedules of HVAC may be provided by Building Management Systems (BMS). Collecting different data from various sources manually can be an error-prone and time-consuming job. Besides, inputting the data into the energy simulation engine is usually a manual process, which requires expertise in BEM. Consequently, both an efficient data model for data integration from multiple information sources and automatic BEM model generation are needed for building energy management.

Ontology is a formal metadata model of semantic web technology that can be used to represent knowledge as a set of concepts within specific domains and relationships among the concepts [8], and could be a solution to data integration for improving the availability and efficiency of information [9]. Recent studies [10–14] have shown that ontology-based approaches are practical and efficient in building energy management with advantages including structured data schema of knowledge representation, convenient bridging for data interoperability and cross-domain linking, and efficient logic inference for implicit

* Corresponding authors.

E-mail addresses: cejcheng@ust.hk (J.C.P. Cheng), cezhewang@ust.hk (Z. Wang).

<https://doi.org/10.1016/j.enbuild.2023.113267>

Received 17 April 2023; Received in revised form 29 May 2023; Accepted 12 June 2023

Available online 16 June 2023

0378-7788/© 2023 Elsevier B.V. All rights reserved.

Nomenclature

AHU	Air Handling Unit
AMY	Actual Meteorological Year
BEM	Building Energy Modeling
BIM	Building Information Modeling
BMS	Building Management System
BOT	Building Topology Ontology
CityGML	City Geography Markup Language
CSV	Comma-Separated Values
EPW	EnergyPlus Weather Format
gbXML	Green Building XML

HVAC	Heating, Ventilation, and Air Conditioning
IDF	EnergyPlus Input File
IFC	Industry Foundation Classes
PMV	Predicted Mean Vote
SHGC	Solar Heat Gain Coefficient
SOSA	Sensor Observation Sampling Actuator
SPARQL	SPARQL Protocol and RDF Query Language
SQL	Structured Query Language
SWRL	Semantic Web Rule Language
TMY	Typical Meteorological Year
VAV	Variable Air Volume

knowledge discovery. Consequently, ontology-based approaches could provide solution to data integration and automatic modeling for BEM. However, ontology-based BEM is less studied and only the ontology-based SimModel [15] has been proposed for building energy simulation purposes but important elements for energy simulation, e.g., materials, space types, thermal zones and space loads, are not considered [6,16]. Hence, further studies of ontology-based approaches should be conducted.

In this study, we propose an ontology-based automatic framework which can integrate data from different sources and generate BEM models with thermal zoning automatically. The following sections in this paper are organized as follows. In Section 2, previous studies related to BIM-BEM and ontology-based building energy management are presented and analyzed. In Section 3, the new automatic framework including ontology models, thermal zoning and model translation methods are proposed. In Section 4, the proposed framework is implemented in a real case to illustrate how the framework works practically. In addition, results of the illustrative example and discussion are given. In Section 5, the main conclusions and findings of this study are summarized.

2. Literature review

2.1. BIM-based BEM

Conversion from BIM to BEM (BIM-BEM) is a hot research topic and has attracted much research attention. The key issue in BIM-BEM studies is to solve the interoperability between BIM and BEM, i.e., to transfer BIM data to a readable file for a BEM Tool (e.g., EnergyPlus Input File (IDF)) [17]. Industry Foundation Classes (IFC) [18] and Green Building XML (gbXML) [19] are the two widely accepted open-sourced schemas of BIM that are commonly used in BIM-BEM. Gao et al. [6] classified BIM-BEM process into 6 steps: geometry (Step 1); construction and materials (Step 2); building or space types (Step 3); thermal zones (Step 4); space loads (Step 5); and HVAC system and its components (Step 6). Previous BIM-BEM studies focused on issues throughout these 6 steps, including incorrect geometry translation, lack of material libraries, lack of space load, lack of HVAC transformation, etc. By geometric boundary extraction [20–25], enhancing IFC or gbXML schemas in energy domains [26–28], supplementing material data [29,30], creating graphical user interface to enable manual addition and modification (e.g., HVAC equipment [31,32]), etc., IFC- or gbXML-based studies can partly address the issues mentioned above. According to Gao et al.'s review [6], the IFC-based studies can achieve Step 2 on average, while gbXML-based studies perform better and can achieve Step 4 on average. Although several studies [31,33,34] can fulfill the whole 6 steps, two issues which hinder BIM-BEM could be identified:

(1) **Dynamic data conversion.** Building data can be categorized into static and dynamic data. Static data types describe physical building information, and dynamic data types involve time-series data [35]. The

current studies mainly focus on automatic BEM during the design phase in which the data are usually static (e.g., geometry, construction and materials). The static data can be translated into building energy models by schema mapping. ASHRAE Research Project RP-1468 [36] provides a compendium of rules for mapping given building features in BIM into BES. For the operation phase, dynamic data from multiple sources (e.g., equipment schedules in time series from BMS) are involved. Current schemas are not mature enough to represent these dynamic data (e.g., complex schedules [33]). In current BIM-BEM studies, schedules were ignored [31] or schedules need to be input manually [37].

(2) **Thermal zoning.** Thermal zoning is a critical step to convert BIM to BEM because different zoning strategies have significant impact on simulation results [38]. Although currently there is no universally-accepted methodology for thermal zoning and methods vary in different standards or guidelines [39–41], Shin et al.'s review paper [42] listed some common criteria for thermal zoning. First, the spaces should be separated into core and perimeter thermal zones and divided by orientation. Further, the spaces with similar attributes (e.g., solar gain, orientation, occupancy, schedule, space function, etc.) should be aggregated into one thermal zone. The properties of spaces may change during the design or operation phase, and the thermal zoning results may change subsequently. Hence, thermal zoning is mutable and can be a dynamic process. Manual thermal zoning hinders the timely BEM and therefore automatic thermal zoning is needed. Furthermore, a static schema cannot achieve the dynamic process of thermal zoning. Current BIM-BEM studies use simplified thermal zones or one-space-to-one-thermal-zone mapping, and in many cases, the preparation of thermal zones are still tedious and time-consuming [6]. Few studies focus on the automation of thermal zoning. Shin and Haberl [38] proposed a procedure for automating thermal zoning based on grid-to-cluster method. However, the preparation of data which support thermal zoning (e.g., space types, setpoints and schedules, which may come from cross-domain information sources) was not considered. Some studies [20,43–46] proposed methods to define geometric data based on IFC second-level space boundaries which can be used for thermal zoning. They focused on algorithms to generate geometric boundaries in a more precise and efficient way, instead of how to generate thermal zoning based on rules suggested by standards or guidelines, e.g., [39–41].

2.2. Ontology-based building energy management

Although ontology-based BEM is less studied as mentioned in Section 1, ontology-based methods have been widely used in other fields of building energy management, from which three key advantages of ontology-based approaches could be observed:

(1) **Structured data schema of knowledge representation.** Entities and their relationships involved in building energy management can be well organized by ontology-based models in a formal metadata format. By using formal data schemas, ontology-based models allow reuse of existing models, which reduces the effort of re-establishment of the

existing contents and enhances the extendibility of the proposed models. Zhang et al. [10] integrated existing ontology models (e.g., Building Topology Ontology (BOT) and Sensor Observation Sampling Actuator (SOSA)) and proposed ontology models to link data across BIM and sensor networks of actual buildings. An ontology of performance indicators was subsequently developed to establish the indicator formulae. With the ontology framework, the performance indicators to support building energy evaluation can be calculated automatically.

(2) Efficient bridging for data interoperability and cross-domain linking. Building energy management involves information from multiple heterogeneous information sources e.g., buildings, sensors, and equipment. Ontology models can serve as bridging media across various domains and create well-connected structures for corresponding information. Corry et al. [11] developed an ontology framework for performance assessment (including building function, thermal loads, and system performance) by integrating data from heterogeneous sources (including building silo, legislation silo, human resources silo, and inventory silo). The ontology framework was implemented in an actual building and took Predicted Mean Vote (PMV) as the example. Results showed that PMV could be calculated via the ontology framework by retrieving data from sensors and BMS. Hu et al. [12] proposed an ontology framework for building energy performance assessment by integrating data from multiple domains (including building silo, calendar silo, weather silo, sensor silo, and indicator silo). The framework was implemented in a school building and results of energy consumption and thermal comfort could be evaluated by cross-domain reasoning using data from domains mentioned above.

(3) Logic inference for implicit knowledge discovery. Implicit knowledge of building energy management, e.g., energy saving measures, are not usually stated in the existing ontology models that contain known knowledge (e.g., physical building information and sensing data). Reasoning, a process of deducing new conclusions from known knowledge, can be conducted based on the ontology models by rule-based or learning-based methods to reveal the implicit knowledge [47]. Han et al. [13] proposed an ontology model to provide a structural framework for organizing information, and to represent knowledge as a set of concepts within a building energy domain. The energy waste context and the energy saving measures could be deduced by rule-based reasoning from the ontology model. The ontology model and reasoning were implemented in an office building. Results showed that energy consumption was reduced by 22.4% compared with the case without reasoning. Degha et al. [14] proposed an ontology-based system which used smart building ontology to organize smart building knowledge and utilized a context-awareness mechanism to provide contextual information by reasoning. Results showed that 40% of total energy consumption could be reduced by implementation of the system compared with the case without implementation.

With these advantages listed above, ontology may be a solution to improving data integration and automatic modeling for BEM by addressing the issues which hinder BIM-BEM as mentioned in Section 2.1. Firstly, Ontology models could serve as a structured data schema of knowledge representation in BEM, which facilitates data preparation for thermal zoning. Secondly, the required data for BEM (including dynamic data) from heterogeneous information sources could be integrated by ontology models. Thirdly, thermal zoning could be conducted by reasoning based on ontology models with inference rules complied with suggested criteria by standards or guidelines [39–41].

In summary, currently, two issues which hinder BIM-BEM process can be identified by the literature review, i.e., thermal zoning and dynamic data conversion. Ontology-based methods could be a solution to remove these hindrances. The purpose of this study is to provide an ontology-based automatic framework for BEM. Our specific goals are: 1) to propose an ontology model that can integrate BEM data from multiple sources including weather, building, internal heat gain, and HVAC system; 2) to propose a cross-domain and rule-based method for thermal zoning based on reasoning of the proposed ontology model; 3) to

propose translation algorithm including dynamic data conversion from the ontology model to the BEM model; 4) to evaluate the performance of the proposed framework in terms of modeling precision and time.

3. Methodology

Fig. 1 shows the overview of the ontology-based automatic framework for BEM. The establishment of the proposed framework includes 3 steps. In step 1, the ontology model is designed, including weather, building, internal heat gain, and HVAC system, which can integrate data from various information sources e.g., weather station, BIM, and BMS. A detailed description of step 1 is given in Section 3.1. In step 2, the inference rule for thermal zoning is set, which can supplement the ontology model with thermal zone entities and relationships. A detailed description of step 2 is given in Section 3.2. In step 3, the automatic translation from the ontology model to the BEM model is established. A detailed description of step 3 is given in Section 3.3.

3.1. Ontology model

There are four key information domains for BEM, weather, building, internal heat gain and HVAC system. Accordingly, four ontology components are designed and constitute the whole ontology model of BEM. Fig. 2 shows the overall framework of the ontology model. The details of each component are illustrated in the following sections and the graphs of the ontology model can be found in Appendix A. The Design Principles of the proposed ontology model include:

(1) Reuse of existing ontology models with necessary extension. To reduce the effort of re-establishment and enhance the extendibility of the proposed models, existing ontology models including Brick Schema (ontology prefix: brick) [48] and BOT (ontology prefix: bot) [49] are introduced. Brick Schema describes HVAC systems comprehensively and BOT focuses on topological relationships among building elements. Meanwhile, necessary extension of the existing ontology models is made to fully support BEM. Relationships between classes from different ontology models are set as the connections among different domains. The new entities and properties of the proposed ontology have a prefix *bes*.

(2) Domain-specific design. The goal of the ontology framework is to integrate data from various information sources, and facilitate the automatic data preparation and building energy modeling. Hence, in the newly proposed ontology model or the existing ontology models, entities and properties are set or introduced according to the indispensable inputs of the energy simulation engines. The entities and properties can be translated to the BEM model directly. No redundant ontology components outside the BEM domain are involved.

(3) Linked with dynamic data. BEM in the operation phase involves data in time series (hereinafter referred to as dynamic data according to Ref. [35]), including weather data and schedules of equipment, occupancy, and HVAC system setpoints. The dynamic data are not integrated into the ontology model. Instead, the dynamic data are stored at their original addresses and in their native format (e.g., databases), while the ontology model provides information (e.g., storage address) which links the dynamic data in which they are stored. Dynamic data for BEM in the operation phase can be accordingly retrieved and translated into BEM models by information from the ontology model and databases.

3.1.1. Weather ontology

There are some existing ontology models of the weather for building energy performance assessment [12] and smart buildings [14]. In these ontology models, each environmental parameter (e.g., temperature and humidity) is set as an individual class. For BEM, weather data are imported into energy simulation engine as one holistic weather file, e.g., EnergyPlus Weather Format (EPW) file. The key information is the addresses where the weather data are stored. According to the Design Principle (3), weather data are dynamic and should be stored at their

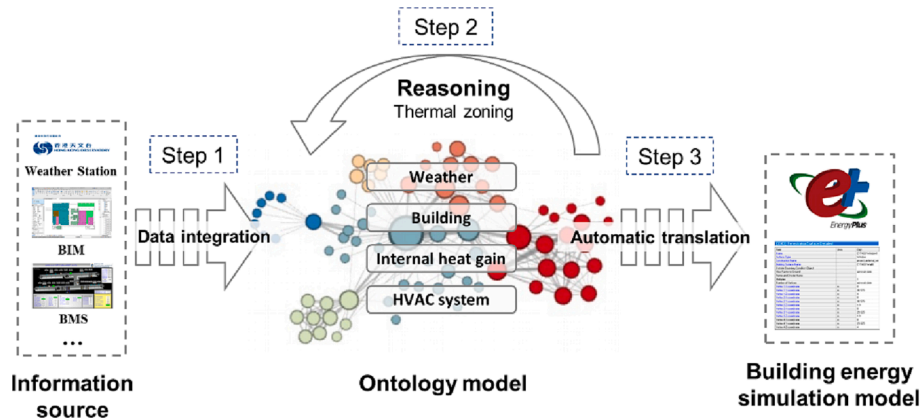


Fig. 1. Overview of the ontology-based automatic framework for BEM.

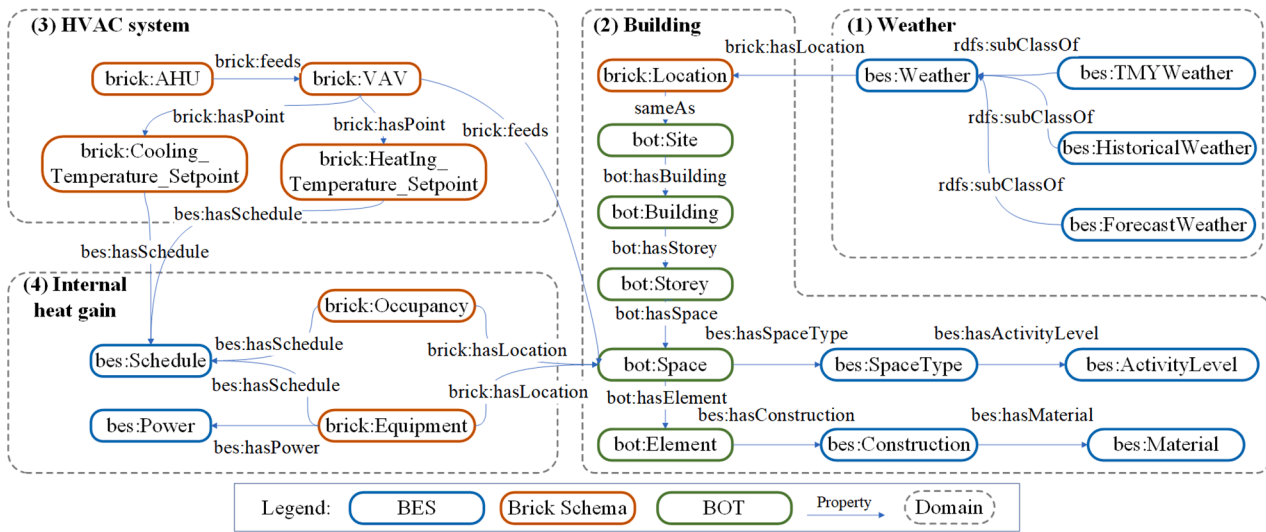


Fig. 2. Overall ontology model framework.

original addresses. Consequently, in this study, weather files are set as classes of weather data types with address information. Fig. A. 1 shows the weather ontology model. According to the type of weather data for BEM, three classes are set, i.e., TMYWeather, HistoricalWeather and ForecastWeather. TMYWeather refers to weather data of a Typical Meteorological Year (TMY), HistoricalWeather refers to historical weather data (e.g., Actual Meteorological Year (AMY)), and ForecastWeather refers to forecast weather data. For forecast weather data, algorithms and hyperparameters are informed by the property `bes:hasForecastMethod`. Each class is equipped with a data property `bes:fileStoreAt` which stores the file addresses for TMY files or databases of TMY data, historical weather data and forecast weather data.

3.1.2. Building ontology

The geometries and thermal properties of construction materials are two key types of building data for BEM. Existing ontology models which describe building information include ifcOWL [50], BOT [49], etc. The ifcOWL model contains detailed and comprehensive building information, whereas BOT focuses on topological relationships among building elements. Although ifcOWL is comprehensive, it contains redundant information for BEM (e.g., `IfcConstructionMgmtDomain` for the construction management domain). BOT is lightweight and flexible for extension, and hence it complies with the Design Principle (2). Consequently, in this study, BOT is adapted to describe building information (Fig. A. 2). Nevertheless, BOT does not contain thermal properties of construction

materials and extensions are proposed.

The BOT extension needs to store the necessary data which are required by energy simulation engines. The necessary data include building's shapes, orientations, thermal properties of walls, roofs, and floors, and fenestration (i.e., windows and openings), which affect the amount of solar radiation that enters the building, the amount of heat that is lost through the building envelope, and the distribution of internal heat gains. The `bot:Space` (Fig. A. 3) describes a geometric unit which contains the data of ceiling height, floor area, etc. These data are set as data properties of the class `bot:Space`. The `bot:Element` describes a construction entity including building surfaces (Fig. A. 4), e.g., wall, floor and roof, and fenestration surfaces (Fig. A. 5), e.g., window. The `bot:Element` contains data including surface type, outside boundary condition and vertex coordinates. Each element should have a type of construction which consists of several layers of materials. Each material is set by necessary thermal properties, i.e., thermal conductivity, specific heat, density, etc. for opaque constructions, and U-factor, Solar Heat Gain Coefficient (SHGC), etc. for windows.

3.1.3. Internal heat gain ontology

Equipment, lighting and occupancy are major sources of internal heat gain. Equipment, lighting and occupancy are located in certain spaces and hence the class `brick:Equipment`, `brick:Lighting` and `brick:Occupancy` are linked to `bot:Space` with object property `brick:hasLocation`. In this study, sensible heat gain is focused on. Internal heat gain can

be quantified using two types of data, i.e., power and schedule. Taking occupancy (Fig. A. 8) as an example, power is related to occupants' activities and is a function of the space type. Schedule is a timetable to indicate runtime of equipment or occupied time of spaces. The timetables are stored in databases and the address information is stored in the ontology model according to the Design Principle (3). In addition, the information sources of the equipment and lighting are indicated by the property `bes:hasSource`.

3.1.4. HVAC system ontology

Brick Schema is an open-source standardized ontology model for physical, logical and virtual assets in buildings and the relationships between them [48]. Brick Schema describes HVAC systems comprehensively, covering sensors, subsystems (e.g., chillers, Air Handling Units (AHU) and Variable Air Volume (VAV) systems) and the relationships between them. The details of the HVAC system ontology model can be found in Ref. [51]. Hence, this study reuses Brick Schema as the ontology model of HVAC systems. Based on the existing ontology model of Brick Schema, extension on schedule of HVAC runtime has been made, which could link HVAC components to databases where the dynamic data are stored according to Design Principle (3). Fig. A. 9 shows an example of VAV temperature setpoints and their schedules in an AHU-VAV system. Each VAV temperature setpoint has a property `bes:hasSchedule` which informs the schedule information and each schedule is linked to the database by the property `bes:fileStoreAt`.

3.2. Thermal zoning

3.2.1. Zoning criteria

To illustrate how the automatic thermal zoning method achieve cross-domain and rule-based thermal zoning based on reasoning of the ontology model, we take one aggregation rule with four criteria to cluster spaces into thermal zone groups for example in this study: Two spaces can be grouped into one thermal zone if.

- (I) they are adjacent to each other,
- (II) they have the same space type,
- (III) they share the same setpoint and schedule, and
- (VI) they are in the same internal area or perimeter with same orientation.

Criterion (II) is related to internal heat gain factor. If two spaces are of the same type, they usually have the similar occupancy density and equipment level. For example, according to Guidelines on Performance-based Building Energy Code provided by The Electrical and Mechanical Services Department of The Government of the Hong Kong Special Administrative Region [52], one value of occupancy density (also applied to lighting power density and equipment power density) is adopted in the same space type. Criterion (III) is related to the control factor. If two adjacent spaces share the same cooling or heating setpoint, they are considered as under the same HVAC control condition. Criterion (VI) is related to the external heat gain factors. Internal areas should be separated from perimeter areas according to Refs. [39–41]. For perimeter spaces, external heat gains differ from solar radiation which varies by orientations [53]. Hence, perimeter spaces with different numbers and directions of external walls should be separated from each other.

The rule adopted by this study follows the common criteria in Ref. [42] that the spaces should be separated into core and perimeter thermal zones and divided by orientation, and the spaces with similar attributes (e.g., solar gain, orientation, occupancy, schedule, space function, etc.) should be aggregated into one thermal zone. It should be noted that the ontology-based thermal zoning method is not limited by the rule and criteria adopted by this study. The method could also be applied to other rules (e.g., the grid/cluster method proposed by Ref. [38]) and future work for the application could be conducted.

3.2.2. Assignment of property `bes:hasExternalWall`

Criteria (I)–(III) can be judged directly according to the entities and relationships in the ontology model proposed in Section 3.1. To facilitate the judgement of Criterion (VI), the data property `bes:hasExternalWall`, is assigned to each space to indicate whether the space has external walls and what the orientations of the external walls are. The assignment of `bes:hasExternalWall` involves vector calculation and is beyond the current ability of reasoning with rule-based language, e.g., vector calculation cannot be achieved by built-ins of Semantic Web Rule Language (SWRL) [54]. Hence, this study processes the assignment by Python scripts, in which the original information of the ontology model is parsed and the generated properties are added to the ontology model.

The assignment of `bes:hasExternalWall` consists of three steps:

(1) Orientation definition. The first step is to determine which orientations are used to describe the external walls. The orientation definition depends on the shape of the building. Fig. 3 shows two examples of orientation definition. Building 1 is of rectangular shape and four orientations are adopted, while Building 2 is in a more complex shape and eight orientations are adopted accordingly. Each orientation is defined with a range (the areas between two dot lines in Fig. 3), and the walls whose directions fall within the range are assigned to the corresponding orientation.

(2) Wall direction identification. Firstly, the information of walls and their vertex coordinates is retrieved from the proposed ontology model. Secondly, the normal vector of the wall surface is calculated. It should be noted that the y-axis of the building coordinate is not always directed to the true north. The normal vector of the wall surface should be checked first and rotated according to the y-axis's degree from the true north if the y-axis of the building coordinate is not directed to the true north. Thirdly, the angles between the normal vector and the orientation vectors are calculated. Finally, the direction of the wall is identified as the orientation with the minimum angle.

(3) Property assignment. Each space is assignment with the data property `bes:hasExternalWall` based on the result of the wall direction identification. For example, `bes:hasExternalWall` of a side space which has one east external wall is denoted with a value "E". The `bes:hasExternalWall` of a corner space which has one north external wall and one east external wall is denoted with a value "N_E". `bes:hasExternalWall` of a space in interior area without external walls (e.g., core spaces in core-perimeter buildings) is denoted with a value "none".

3.2.3. Reasoning of thermal zoning

After assignment of `bes:hasExternalWall`, all properties to support the judgement of thermal zoning criteria are fully prepared. The rule-based language SWRL [54] is applied to enable reasoning for thermal zoning. Table 1 shows the inference rule for thermal zoning written in

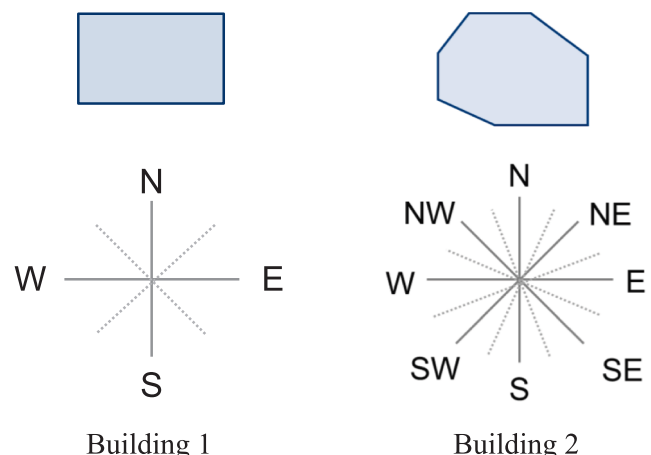


Fig. 3. Examples of orientation definition.

Table 1
Reasoning for thermal zoning.

Reasoning 1: Thermal zoning	
1	bot:Space(?spacea)
2	^bot:Space(?spaceb)
3	^bot:adjacentZone(?spacea,?spaceb)
4	^bes:hasSpaceType(?spacea,?spacetypea)
5	^bes:hasSpaceType(?spaceb,?spacetypeb)
6	^sameAs(?spacetypea,?spacetypeb)
7	^brick:feeds(?terminala,?spacea)
8	^brick:Cooling_Temperature_Setpoint(?setpointa)
9	^brick:isPointOf(?setpointa,?terminala)
10	^bes:Schedule(?schedulea)
11	^bes:hasSchedule(?setpointa,?schedulea)
12	^brick:feeds(?terminalb,?spaceb)
13	^brick:Cooling_Temperature_Setpoint(?setpointb)
14	^brick:isPointOf(?setpointb,?terminalb)
15	^bes:Schedule(?scheduleb)
16	^bes:hasSchedule(?setpointb,?scheduleb)
17	^sameAs(?schedulea,?scheduleb)
18	^bes:hasExternalWall(?spacea,?walllista)
19	^bes:hasExternalWall(?spaceb,?walllistb)
20	^swrlb:equal(?walllista,?walllistb)
21	->bes:sameThermalZone(?spacea,?spaceb)

SWRL. First, two spaces are checked whether they are adjacent to each other (Criterion (I)) by property bot:adjacentZone (Lines 1–3). Secondly, the two spaces are checked whether they have the same space type (Criterion (II)) by comparing the values of the property bes:hasSpaceType (lines 4–6). Thirdly, the two spaces are checked whether they share the same setpoint and schedule (Criterion (III)) according to the setpoints in the two spaces. The reasoning rules need to reveal the setpoints belonging to the two spaces according to serving relationships between the VAVs and spaces (lines 7–17). Finally, the two spaces are checked whether they have similar external heat gain (Criterion (IV)) by comparing the values of the property bes:hasExternalWall (lines 18–20). After reasoning, spaces in the same thermal zone are assigned an object property bes:sameThermalZone to relate each other.

3.3. Model translation

The third step of the ontology-based framework for BEM is the establishment of automatic translation from the ontology model to the BEM model. To achieve accurate and seamless translation, three issues should be addressed:

(1) Field mapping. Representations of entities and properties are different between the ontology model and the BEM model. The first issue to achieve the accurate mapping of the same entities and properties in different models. It is a common issue in BIM-BEM studies (e.g., space object mapping between IFC and IDF in Ref. [55]), but it is less studied in ontology-to-BEM.

(2) Dynamic data conversion. In the operation phase, dynamic data from multiple sources are involved. For simulations in different periods, different ranges of dynamic data should be retrieved and written in the configuration files of the energy simulation engine. How to transfer the dynamic data via the ontology model into the BEM model is the second issue to be solved.

(3) Completeness of configuration files. Besides inputs of weather, building, internal heat gain and HVAC system, there are some specific fields, e.g., Simulation Control, in configuration files. How to complete the configuration files to run the simulation successfully is the third issue to be solved.

Accordingly, methods are proposed to address the three issues listed above in the following sections. In this study, the widely-accepted simulation tool EnergyPlus [4] is taken as the example energy simulation engine to illustrate the methodology of model translation.

3.3.1. Instance-based mapping

The Instance-based method maps corresponding fields individually between two models in different schemas. The method was adopted in previous studies [56–58]. For example, Deng et al. [58] applied instance-based method to generate mapping rules between IFC and City Geography Markup Language (CityGML). Results showed that a wider range of geometry information could be captured compared to other methods [59–61]. Objects are written separately as individual items (Fig. 4) in both the ontology file (e.g., OWL file in Turtle syntax [62]) and the energy simulation configuration file (e.g., IDF). This feature facilitates the implementation of the instance-based method for field mapping. Consequently, the instance-based method is adopted to address the field mapping issue in this study.

There are two levels of field mapping. The first level is entity mapping, in which the corresponding entities are identified between the two models (e.g., CYT4001Occupancy in Fig. 4). The second level is property mapping, in which the corresponding properties are identified under the same entity between the two models (e.g., occupancy name, space where the occupancy is located, etc. in Fig. 4). The “must-have” inputs into the model are determined by the objectives of simulations. Appendix B provides the detailed information of entity mapping and property mapping between the proposed ontology model and BEM model for thermal load simulation. Fig. 5 takes the occupancy as the example to clearly show how the properties are translated from the proposed ontology model to the BEM model (Fig. 5 (a)). There are three types of property mapping. Some properties (the red arrows in Fig. 5 (a)) can be mapped directly. Some properties (the green arrows in Fig. 5 (a)) should be calculated based on the ontology model and the database. Some properties (e.g., number of people) are set as default values.

Information of spaces and thermal zones from the proposed ontology model is automatically translated into the BEM model based on instance-based mapping (Fig. 5 (b)). The space information is translated into the field *Space* which contains the information which zone the space belongs to. Spaces sharing the property bes:sameThermalZone are aggregated into one zone. Each individual space, which does not have the property bes:sameThermalZone and should not be aggregated with others, is set as one individual zone.

After field mapping, an automatic translation framework from OWL files and IDF files is established based on the Python scripts as shown in Fig. 4. The Python library rdflib [63] is adopted to parse the ontology model and extract the information of the targeted fields via SPARQL Protocol and RDF Query Language (SPARQL) query. The Python library eppy [64] is implemented to write the IDF file based on the extracted information. IDF files in different EnergyPlus [4] versions can be generated by eppy according to the assigned IDD files, which ensures the identification model is extensible to all objects in EnergyPlus [4] and future versions. Example scripts for material information translation are shown in Appendix C. Some items which are “must-have” inputs to BEM models may not be available (especially during the design stage) and miss in the ontology model. The missing items can be identified by the translation scripts and information can be raised subsequently to inform users which items miss. According to the error information, users could supplement the missing items in the ontology model.

3.3.2. Dynamic data conversion

Weather data and schedules of equipment and occupancy are two main dynamic data which should be translated into the BEM model. As mentioned in Section 3.1.1, three kinds of weather data are involved in BEM in the operation phase. TMY weather contains weather data for a typical year and are stored in weather files (e.g., EPW files) which could be directly imported into energy simulation engine. Historical and forecast weather data are continuously added over time. To simulate building energy consumption in one past or future period, corresponding weather data should be selected according to the period and input into the energy simulation engine. Consequently, the conversion of historical and forecast weather data is considered in this study. For

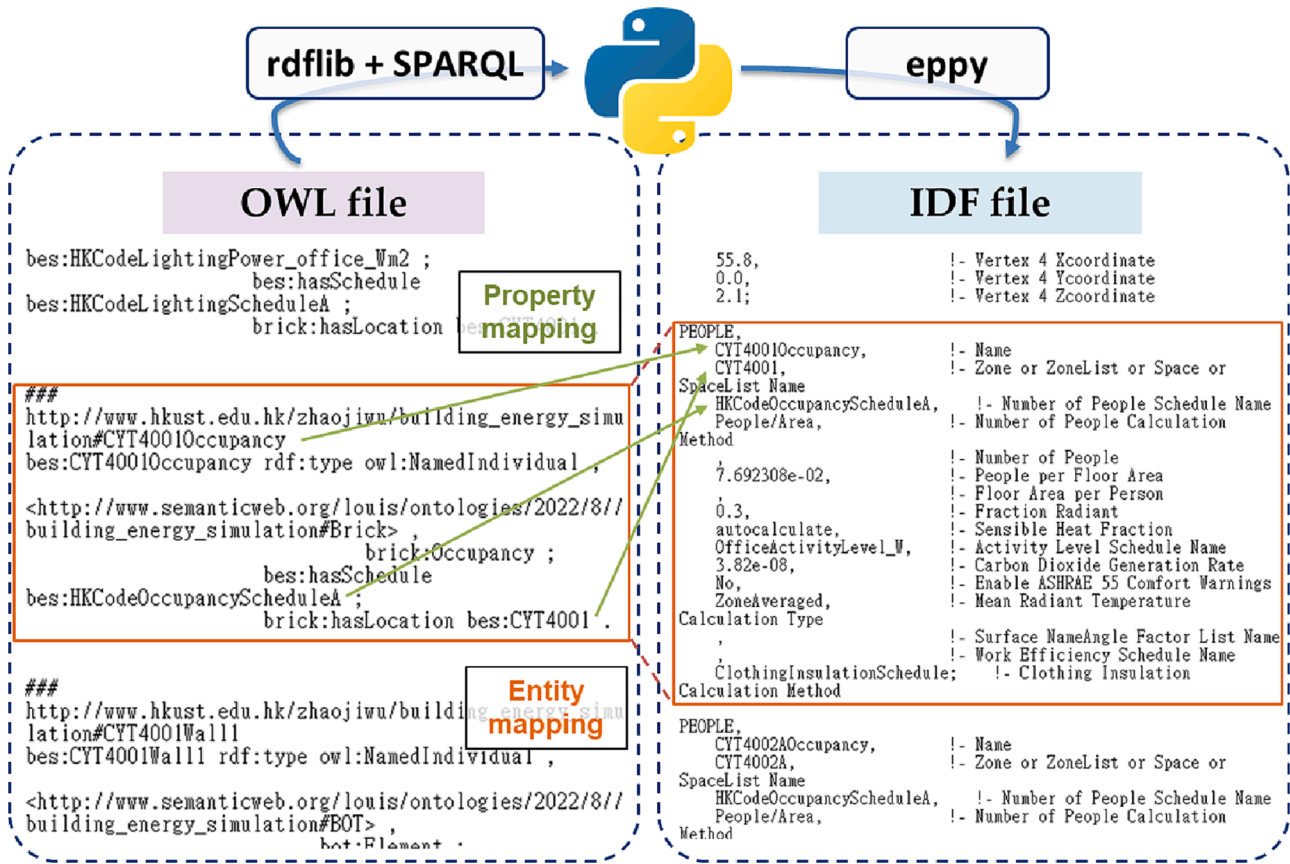


Fig. 4. Instance-based mapping between the ontology model and the BEM model.

schedules of equipment and occupancy, the selection also depends on the simulation period. For the simulation in past periods, the historical records of the occupancy sensors and equipment status should be selected. For the simulation in future periods, since there are no records, the common practice is to use the historical data to represent the future conditions (e.g., to use the average number in one past period).

Fig. 6 shows the overall workflow of dynamic data conversion from the database to the BEM model. According to the Design Principle (3), dynamic data are stored in the original addresses instead of the ontology model and storage address information is provided by the proposed ontology model. The first step is to retrieve the data address from the proposed ontology model via the query language SPARQL. With address information and the simulation period, the query in Structured Query Language (SQL) can be constructed and sent to the databases which store the dynamic data. Finally, data can be retrieved according to the query and translated to EPW or IDF file via Python scripts and the library eppy. To deal with issues related to missing data, erroneous data and anomalous data, data preprocessing should be conducted before generation of weather files and configuration files. Existing imputation techniques for building monitoring data (Linear Interpolation, K-Nearest Neighbor, Multiple Imputations through Chained Equations, etc.) [65] could be implemented.

Table 2 shows the algorithm to translate weather data to the BEM model. The inputs include the simulation period, weather file template and ontology model. The weather file template is the local weather file in EPW format. Recorded historical or forecast weather data, which contain relevant parameters, e.g., temperature and relative humidity, in time series, are retrieved from the database. The columns (parameter values) and the rows (timestamps) in the weather file template are located according to the simulation period. The template data in these columns and rows are covered by the actual data. For missing data, missing values notations (e.g., 999 for direct normal radiation in EPW

file [66]) is written. Finally, the EPW file with the actual data is generated and input into the energy simulation engine.

Table 3 shows the algorithm to translate the equipment schedule to the BEM model. Two scenarios of schedules are studied. Scenario 1 (hereinafter referred to as the record-based scenario): Equipment status (e.g., runtime and percentage of full operation) is recorded by BMS and equipment schedules could be exported from BMS. Scenario 2 (hereinafter referred to as the guideline-based scenario): Records of equipment status are unavailable and the common practice is to use schedules provided by guidelines or standards. In the building energy simulation engine, e.g., EnergyPlus [4], schedules usually consist of hourly data. For the guideline-based scenario, hourly data could be provided by guidelines directly. For the record-based scenario, the equipment status may be recorded in different intervals (e.g., in minute, depending on the setting of BMS). A calculation should be conducted to obtain hourly data, e.g., to use the average of minutely equipment status records in one hour as the representative of this hour.

Table 4 shows the algorithm to translate the occupancy schedule to the BEM model. The occupancy schedule translation is similar to the equipment schedule translation, whereas the occupancy is usually recorded by the number of people and should be translated into the percentages of the maximum occupancy. In the record-based scenario, the fields Number of People and People per Area should be calculated automatically according to records by occupancy sensors.

3.3.3. Template-based configuration file generation

Through instance-based mapping and dynamic data conversion, the information from the proposed ontology model and the databases can be translated into the BEM model automatically. However, to complete the simulation configuration file and run the simulation successfully, some indispensable fields still need to be supplemented. Fig. 7 (a) shows the fields (in dashed frames) which should be supplemented. These fields

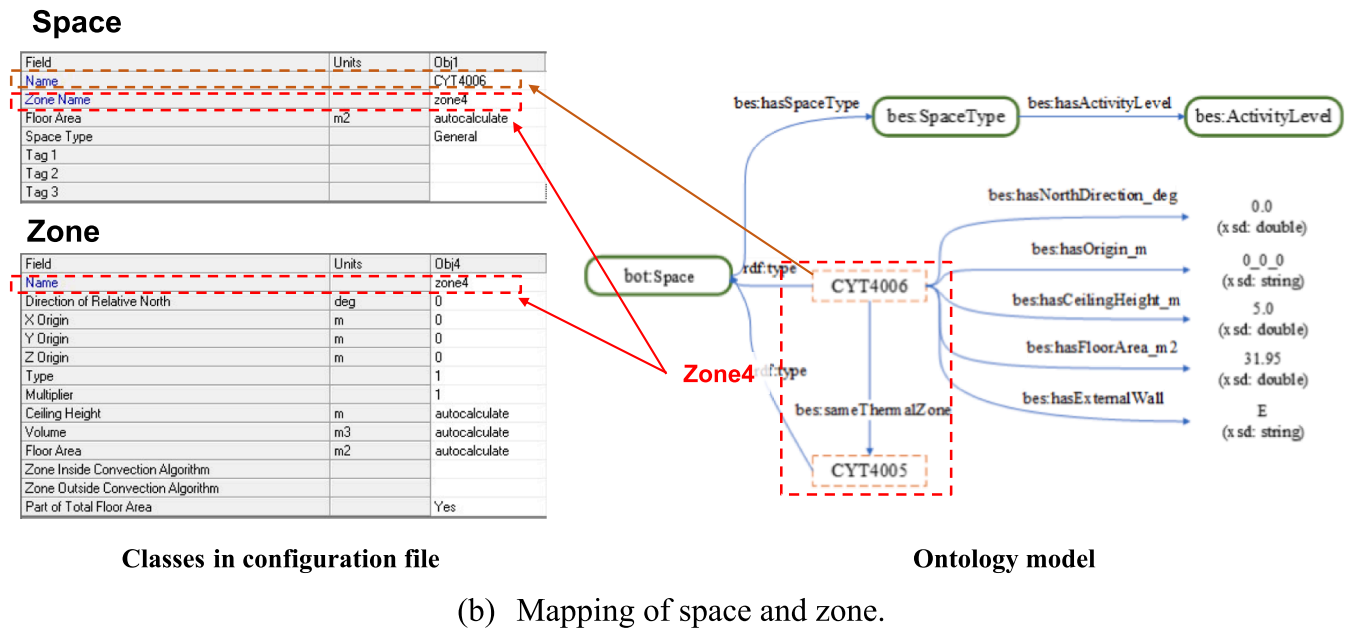
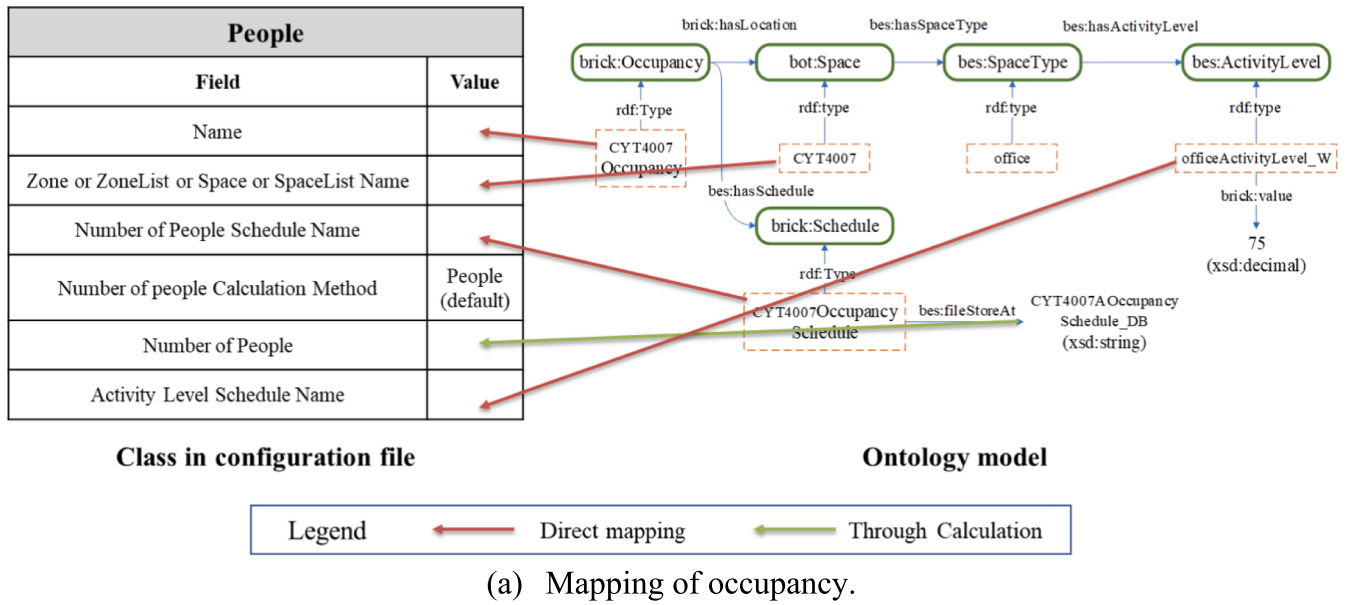


Fig. 5. Instance-based mapping between the ontology model and the BEM model.

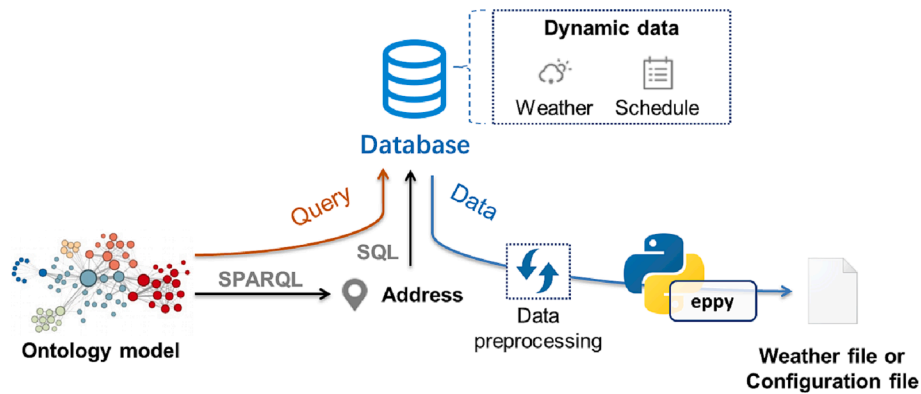


Fig. 6. Dynamic data conversion workflow.

Table 2
Algorithm to translate weather data to the BEM model.

Algorithm 1: Weather file generation	
Input: start timestamp T_s , end timestamp T_e , ontology model O , weather database D , weather file template F	
1	query and retrieve <i>weather_address</i> $\leftarrow O$
2	query and retrieve <i>weather_data</i> [] $\leftarrow D$ according to <i>weather_address</i>
3	preprocess data (Linear Interpolation or K-Nearest Neighbor, etc.)
4	open F do
5	for each <i>weather_parameter</i> in <i>weather_data</i> [] do
6	<i>parameter_data_list</i> [] $\leftarrow$ hourly data from <i>weather_data</i> [] between T_s and T_e
7	locate <i>column</i> of <i>weather_parameter</i> in F
8	locate <i>row</i> between T_s and T_e in F
9	write <i>parameter_data_list</i> [] in <i>column</i> and <i>row</i>
10	end for
11	close F
Output: EPW file F	

Table 3
Algorithm to translate equipment schedule to the BEM model.

Algorithm 2: Lighting / Electrical equipment schedule translation	
Input: start timestamp T_s , end timestamp T_e , ontology model O , schedule file or database S , IDF file F	
1	query and retrieve <i>space_list</i> [] $\leftarrow O$
2	open F do
3	for each <i>space</i> in <i>space_list</i> [] do
4	query and retrieve <i>schedule_address</i> , <i>area</i> , <i>power</i> $\leftarrow O$
5	query and retrieve <i>schedule</i> [] $\leftarrow S$ according to <i>schedule_address</i>
6	if the schedule is guideline-based then
7	<i>equipment_hourly_status_list</i> [] $\leftarrow$ hourly data from <i>schedule</i> []
8	write <i>equipment_hourly_status_list</i> [] to <i>Schedule:Day:Hourly</i> in F
9	<i>level_per_area</i> $\leftarrow$ <i>power</i>
10	write <i>level_per_area</i> to <i>Watts per Zone Floor Area</i> in F
11	if the schedule is record-based then
12	preprocess data (Linear Interpolation or K-Nearest Neighbor, etc.)
13	<i>equipment_status_list</i> [] $\leftarrow$ <i>schedule</i> [] between T_s and T_e
14	for each <i>hour</i> every weekday or weekend do
15	<i>equipment_hourly_status_list</i> [] $\leftarrow$ average(<i>equipment_status_list</i> [] in <i>hour</i>)
16	<i>level_per_area</i> $\leftarrow$ <i>power</i> / <i>area</i>
17	write <i>level_per_area</i> to <i>Watts per Zone Floor Area</i> in IDF file
18	write <i>equipment_hourly_status_list</i> [] to <i>Schedule:Day:Hourly</i> in F
19	end for
20	write <i>Schedule:Week:Daily</i> in F
21	write <i>Schedule:Year</i> in F
22	end for
23	close F
Output: IDF file F	

usually are related to the settings of the simulation, e.g., simulation control, calculation methods and output parameters. For the same simulated object, these settings seldom change in different simulation scenarios. This study adopts a template-based configuration file generation method instead of storing the settings information in the ontology model (which needs extra ontology objects and model translation algorithms) to avoid redundant procedures. Fig. 7 (b) shows the workflow of the method. The fields related to the simulation settings are written in the template configuration file. Other fields are written to the template through instance-based mapping and dynamic data conversion by the Python scripts for model translation. The full configuration file is generated after writing and can be imported into the energy simulation engine.

4. Illustrative example

In this section, a real case was studied to illustrate the implementation of the proposed framework.

Table 4
Algorithm to translate occupancy schedule to the BEM model.

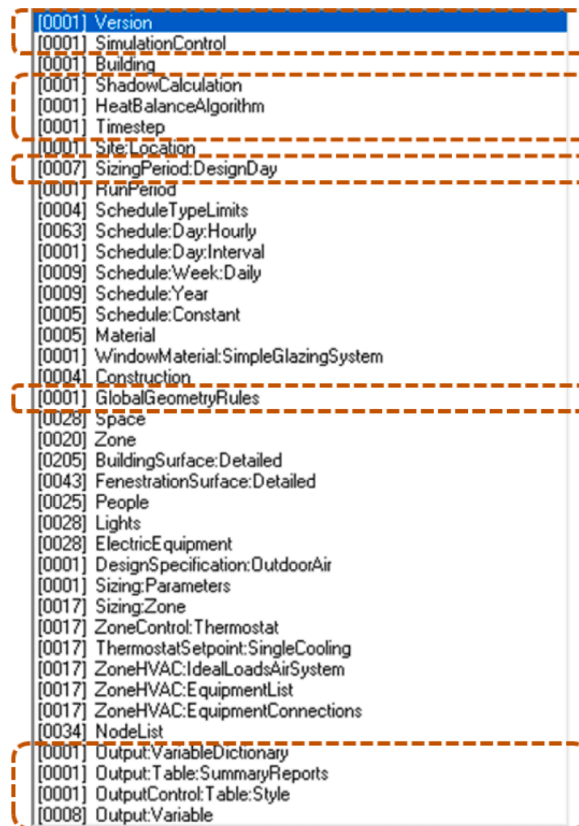
Algorithm 3: Occupancy schedule translation	
Input: start timestamp T_s , end timestamp T_e , ontology model O , schedule file or people counting database S , IDF file F	
1	query and retrieve <i>space_list</i> [] $\leftarrow O$
2	open F do
3	for each <i>space</i> in <i>space_list</i> [] do
4	query and retrieve <i>schedule_address</i> , <i>area</i> $\leftarrow O$
5	query and retrieve <i>schedule</i> [] $\leftarrow S$ according to <i>schedule_address</i>
6	if the schedule is guideline-based then
7	<i>occupancy_hourly_ratio_list</i> [] $\leftarrow$ hourly data from <i>schedule</i> []
8	write <i>occupancy_hourly_ratio_list</i> [] to <i>Schedule:Day:Hourly</i> in F
9	<i>people_per_area</i> $\leftarrow$ people per area given by the guideline
10	write <i>people_per_area</i> to <i>People per Floor Floor Area</i> in F
11	if the schedule is record-based then
12	preprocess data (Linear Interpolation or K-Nearest Neighbor, etc.)
13	<i>occupancy_count_list</i> [] $\leftarrow$ <i>schedule</i> [] between T_s and T_e
14	<i>number_of_people</i> $\leftarrow$ max(<i>occupancy_count_list</i> [])
15	<i>people_per_area</i> $\leftarrow$ <i>number_of_people</i> / <i>area</i>
16	write <i>people_per_area</i> to <i>People per Floor Floor Area</i> in F
17	for each <i>hour</i> every weekday or weekend do
18	<i>occupancy_hourly_ratio</i> $\leftarrow$ average(<i>occupancy_count_list</i> [] in <i>hour</i>) / <i>number_of_people</i>
19	write <i>occupancy_hourly_ratio</i> to <i>Schedule:Day:Hourly</i> in F
20	end for
21	write <i>Schedule:Week:Daily</i> in F
22	write <i>Schedule:Year</i> in F
23	close F
Output: IDF file F	

4.1. Framework implementation

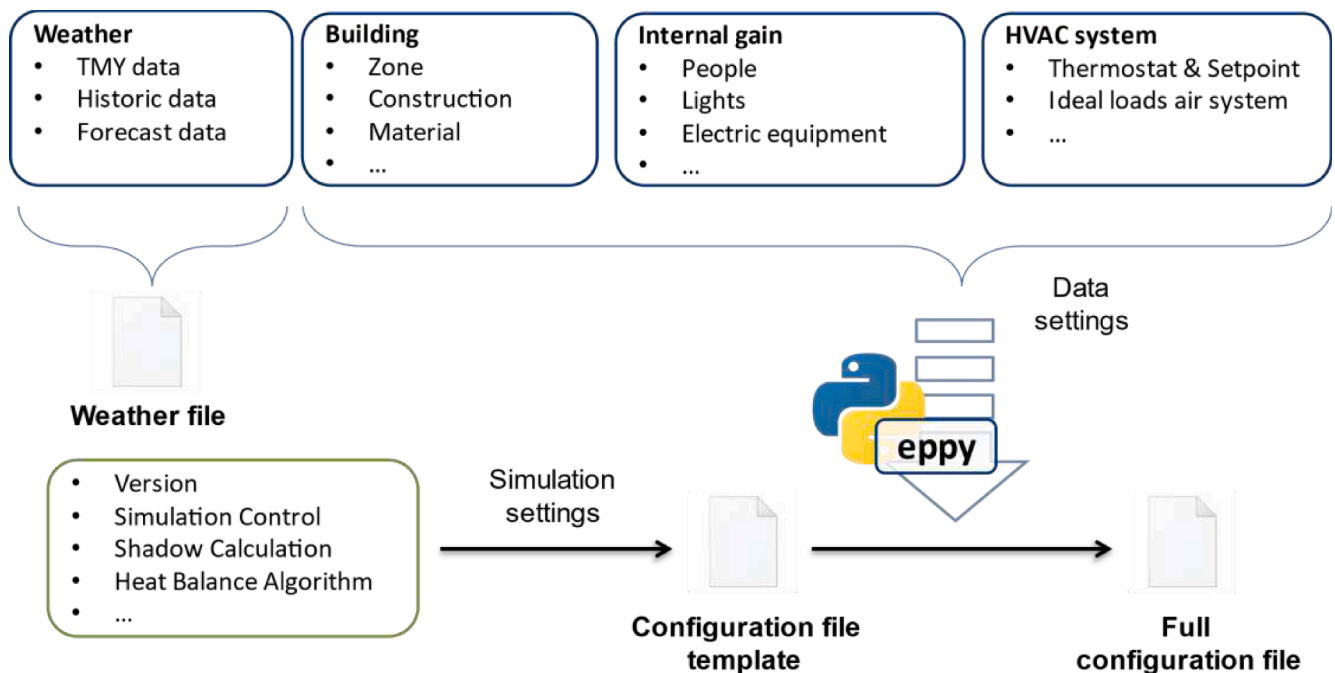
One floor of a campus building of a university in Hong Kong was selected as the case object (Fig. 8). The floor is used for academic purposes with an area of around 2200 m² and contains approximately 30 rooms which can be categorized into offices, laboratories, corridors, toilets, and staircases, with two kinds of HVAC terminals serving the floor. The interior area is served by VAVs and the perimeter area is served by FCUs. In November 2022, the building managers planned to verify the building energy simulation results by EnergyPlus [4] in this floor to support a pilot study on predictive control in the future. It was assumed that the selected floor was adiabatic to the floor above and below. They had collected the field measurement data and planned to conduct building energy simulation for one week (7 November 2022 to 13 November 2022) using EnergyPlus [4].

The proposed framework was implemented to develop the BEM model of this floor. A desktop computer served as the platform and the specifications are listed as follows: Intel Core CPU i7-6700 at 3.40 GHz 8 cores, 32 GB RAM memory, and Windows 64 operating system. The versions of software or tools used are listed as follows: Protégé 5.5.0, Python 3.9.13, and EnergyPlus 9.6. Fig. 9 shows the implementation workflow of the proposed framework in the illustrative example. The implementation was based on the generic methods introduced in Section 3 and the specific settings of the illustrative example are described as follows.

Step 1: Ontology modeling. The ontology model was developed in the software Protégé [67] based on the method described in Section 3.1. Fig. 10 (a) shows the classes, data properties, object properties, parts of the individuals, and metrics of the ontology model in Protégé. For the weather, the TMY data were obtained from the EnergyPlus official website in EPW format [68], the historical weather data came from the local weather station on the campus, and the forecast weather data came from the Hong Kong Observatory [69]. For the building, the Revit [70] BIM models, which contained the information of the geometry and thermal properties of the envelope construction served as the building information sources for the ontology models. For the internal heat gain, the occupancy information (number of occupants in spaces) was



(a) Fields of simulation settings, etc. (in dashed frames), which are not translated from the ontology model.



(b) Workflow of template-based configuration file generation.

Fig. 7. Template-based configuration file generation.

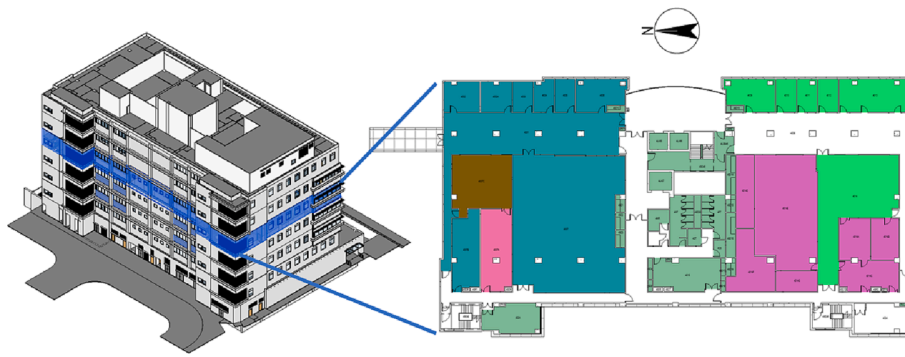


Fig. 8. Location and layout of the case.

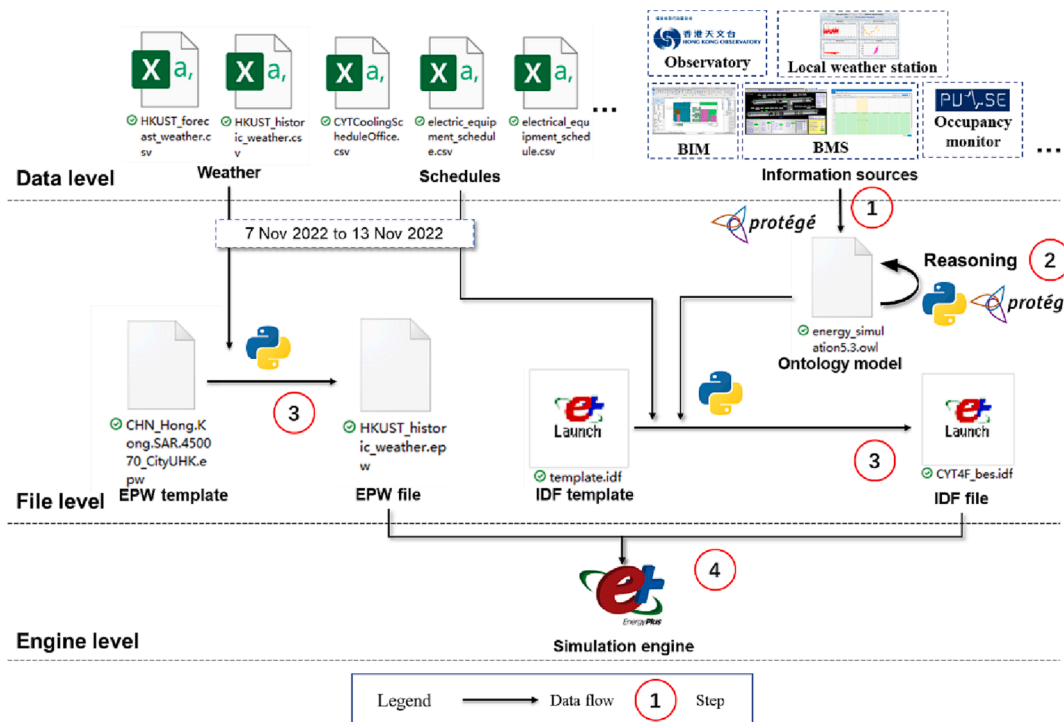


Fig. 9. Workflow of the proposed framework for the illustrative example.

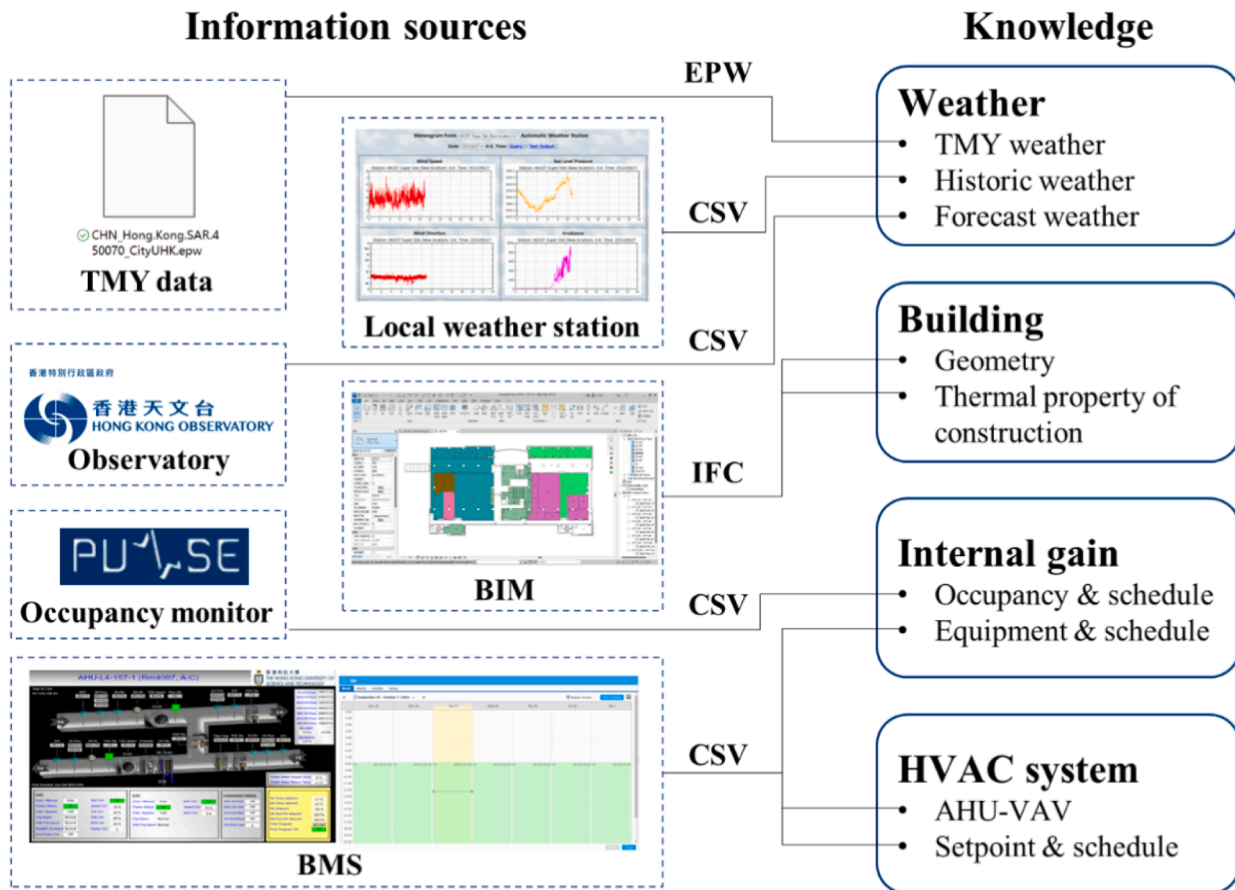
retrieved from a campus project which monitored the number of the facility users. The equipment information (location, power, schedule, etc.) was retrieved from the BMS of the campus (Delta Controls [71]). The BMS also served as the information sources for the HVAC system ontology model. The final ontology model was stored in Turtle syntax as an OWL file [62].

Step 2: Thermal zoning. The building is of rectangular shape and four orientations (i.e., north, west, south, and east) were adopted. The y-axis of the building's BIM model is directed to the east (Fig. 8), and hence the normal vectors of the wall surfaces should be rotated 270° clockwise before wall direction identification. The reasoning for thermal zoning was conducted in Protégé based on SWRL, using the rule listed in Table 1. After Step 2, the ontology model was supplemented with thermal zone information and stored in Turtle syntax as an OWL file [62].

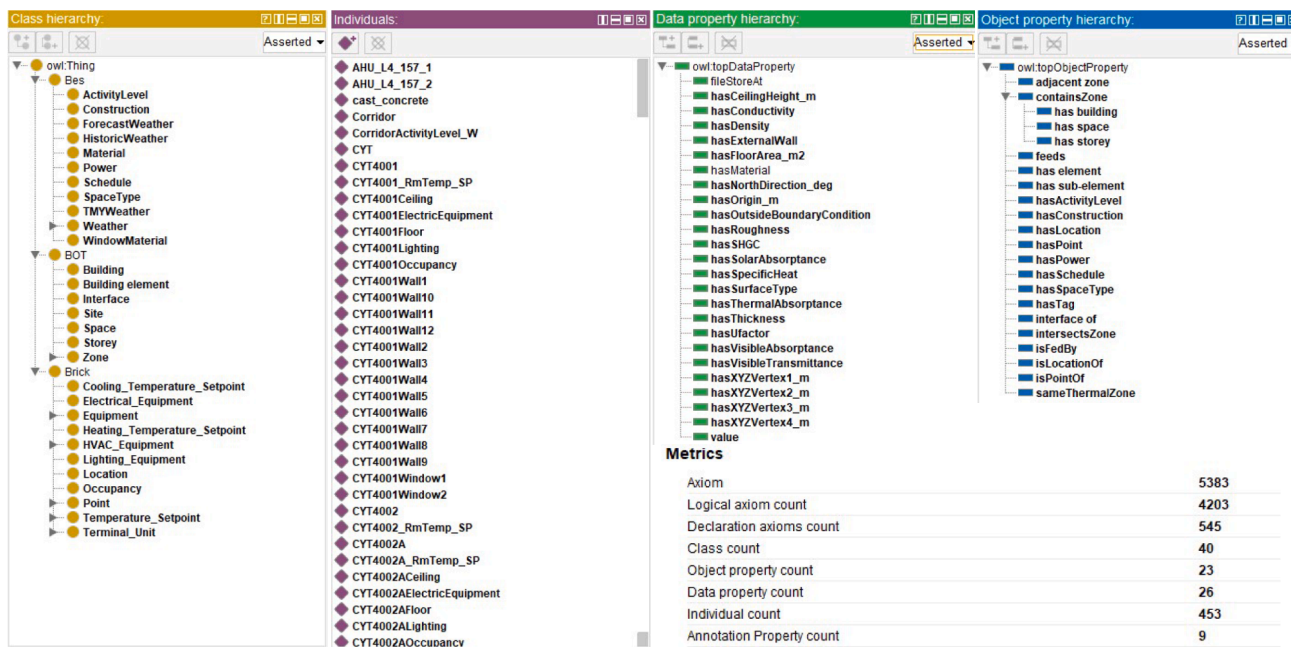
Step 3: Building energy modeling. The weather data from 7 November 2022 to 13 November 2022 were retrieved from the historical records of the campus weather station. The data were downloaded in original Comma-Separated Values (CSV) format with timestamp and value columns, and were checked to ensure there were no missing data, erroneous data and anomalous data. The checking results showed that

no data were in error and no imputation was conducted. The parameters of the station records included dry bulb air temperature, relative humidity, atmospheric pressure, wind direction, and wind speed. For other parameters in a EPW file that cannot be provided by the weather station, the parameters were given the missing value notations [66]. A pilot study had been conducted and results showed that the CV(RMSE) (Eq. (1)) was within 5% if the parameters were given the missing value notations comparing with adopting the hourly average of TMY data. For the spaces with occupancy sensors and records of equipment status from BMS (the record-based scenario), the schedules of occupancy and equipment from 7 November 2022 to 13 November 2022 were retrieved from the record databases and written to the IDF file. The data were downloaded in original CSV format and were checked to ensure there were no missing data, erroneous data and anomalous data. No data were in error and no imputation was conducted. For the spaces without occupancy sensors and records of equipment status from BMS (the guideline-based scenario), schedules from Guidelines on Performance-based Building Energy Code provided by The Electrical and Mechanical Services Department of The Government of the Hong Kong Special Administrative Region [52] were adopted.

Step 4: Building energy simulation. The EPW file and IDF file



(a) Knowledge in BEM and information sources.



(b) Classes, properties, individuals, and metric of the ontology model in Protégé.

Fig. 10. Ontology modeling.

generated by step 3 were imported into the building energy simulation. The run period was set from 7 November 2022 to 13 November 2022. Cooling is still needed during the transition seasons in Hong Kong, located in a hot-humid area. Therefore, the building managers focused on the cooling control during the transition seasons in the further energy-simulation-based predictive studies. The cooling load was the main parameter of the future studies and hourly data of the cooling load were exported from the building energy simulation for further analysis.

To evaluate the performance of the proposed framework, manual modeling was introduced for comparison. Manual modeling is the basic and reliable method that can serve as the benchmark for evaluation of new automatic methods in terms of modeling time and precision. The reliability of manual modeling can be ensured by modelers' domain knowledge and iterative revision. Five BEM modelers, each of whom had 5-to-9-year experience in thermal load simulation by EnergyPlus [4], were invited to manually develop the energy simulation model with the same information as the proposed framework, using IDF Editor provided by EP-Lauch, the official EnergyPlus tool [72]. The participants were not involved in the development of this study and not informed the objectives of this study. The participants were required to: 1) retrieve actual weather data from the campus weather station database, 2) build the geometric model of the floor and extract material data of the constructions according to the BIM model provided by the building managers, 3) retrieve equipment and occupancy information from the campus BMS, 4) conduct thermal zoning, 5) edit the EPW and IDF file, and 6) run the simulation. To ensure the accuracy of the manual modeling, the students checked each other's models after each step (for example, after the retrieval of weather data and the EPW file edition, the EPW files were checked by each other) and finally submitted one BEM model on which they reached a consensus. The overall working time from the start to the submission of the final BEM model was recorded.

4.2. Results

4.2.1. Thermal zoning

Table 5 shows the space types, cooling schedules and external wall directions of the spaces. The space types and cooling schedules were

Table 5
Properties of the spaces.

Space	Space type	Cooling schedule	External wall direction
CYT4001	Office	CYTCoolingScheduleOffice	N
CYT4002	Office	CYTCoolingScheduleOffice	E
CYT4002A	Office	CYTCoolingScheduleOffice	E,N
CYT4003	Office	CYTCoolingScheduleOffice	E
CYT4004	Office	CYTCoolingScheduleOffice	E
CYT4005	Office	CYTCoolingScheduleOffice	E
CYT4006	Office	CYTCoolingScheduleOffice	E,S
CYT4007	Lab	CYTCoolingScheduleLab	none
CYT4007A	Office	CYTCoolingScheduleOffice	none
CYT4007B	Office	CYTCoolingScheduleOffice	none
CYT4007C	Office	CYTCoolingScheduleOffice	none
CYT4008	Office	CYTCoolingScheduleOffice	S
CYT4009	Office	CYTCoolingScheduleOffice	E,N
CYT4010	Office	CYTCoolingScheduleOffice	E
CYT4011	Office	CYTCoolingScheduleOffice	E
CYT4012	Office	CYTCoolingScheduleOffice	E
CYT4013	Office	CYTCoolingScheduleOffice	E,S
CYT4014	Lab	CYTCoolingScheduleLab	none
CYT4014E	Lab	CYTCoolingScheduleLab	none
CYT4015	Office	CYTCoolingScheduleOffice	none
CYTcor1	Corridor	CYTCoolingScheduleCorridor	E
CYTcor2	Corridor	CYTCoolingScheduleCorridor	none
CYTcor3	Corridor	CYTCoolingScheduleCorridor	N,S,W
CYTcor4	Corridor	CYTCoolingScheduleCorridor	N
CYTcor5	Corridor	CYTCoolingScheduleCorridor	S
CYTnoac1	Other	–	none
CYTnoac2	Other	–	N,S,W
CYTnoac3	Other	–	N,S,W

extracted from original information of the ontology model and the external wall directions were obtained after the automatic assignment of property `bes:hasExternalWall` (Section 3.2.2). Fig. 11 (a) shows the layout and the space types of the spaces. The spaces in the illustrative example shared the same cooling setpoint schedule if they belonged to the same space type. Hence, in this case, Criterion (II) of the zoning rule was equivalent to Criterion (III). There were four types of the spaces in the case and the adjacent spaces with the same space type should be grouped into one thermal zone. Take the office cluster in the northeast corner of the floor (upper left corner in Fig. 11 (a)) for example. 4002A, 4003, 4004 and 4005, which were adjacent offices, should be grouped into one thermal zone. However, although 4001, 4002 and 4006 were adjacent to the group (4002A, 4003, 4004 and 4005) and all of them belonged to the office type, they were not grouped together because 4001, 4002, 4006 and the group had different external wall directions. 4001 had one north wall. 4002 had one north wall and one east wall. 4006 had one east wall and one south wall. The group had one east wall. The spaces with walls in different directions received different solar radiation and Criterion (IV) ensured that they should belong to different thermal zones.

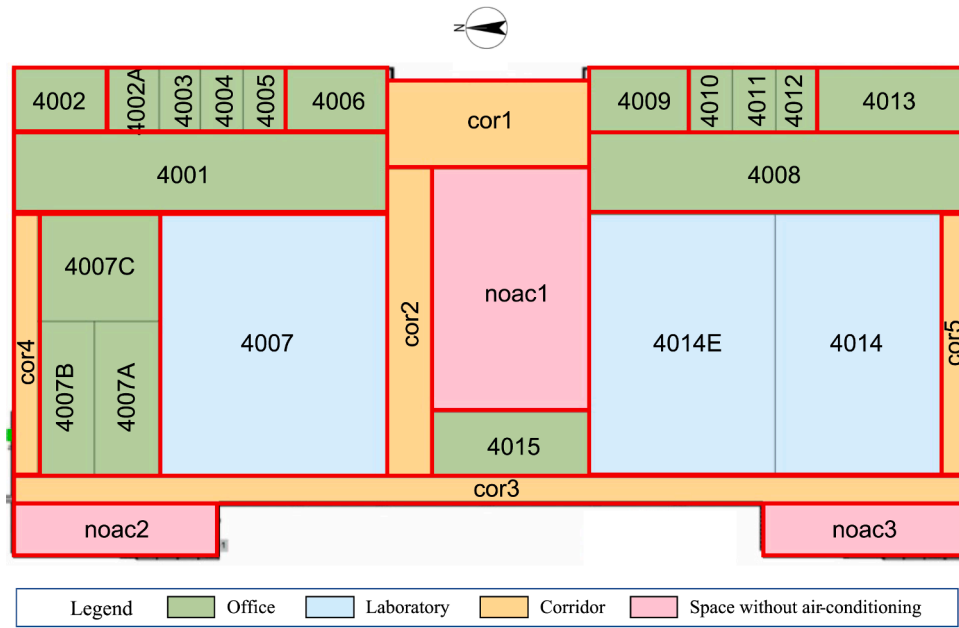
The red lines in Fig. 11 (a) show the expected thermal zoning result. Fig. 11 (b) shows the thermal zoning result by the reasoning. In total, 20 thermal zones were generated. By comparing Fig. 22 (a) and Fig. 22 (b), it could be concluded that the reasoning result kept the same as the expected result. Fig. 11 (b) clearly shows that the floor was divided into the interior and external area. For each area, zones were divided according to different space types and setpoint schedules. For the external area, zones were divided according to external walls in different directions (e.g., the corner space 4002 with external wall direction “N,E” and the size space 4002A with external wall direction “E” belonged to different thermal zones). The proposed thermal zoning method complies with the common zoning rules provided by standards or guidelines [39–41].

4.2.2. Building energy simulation

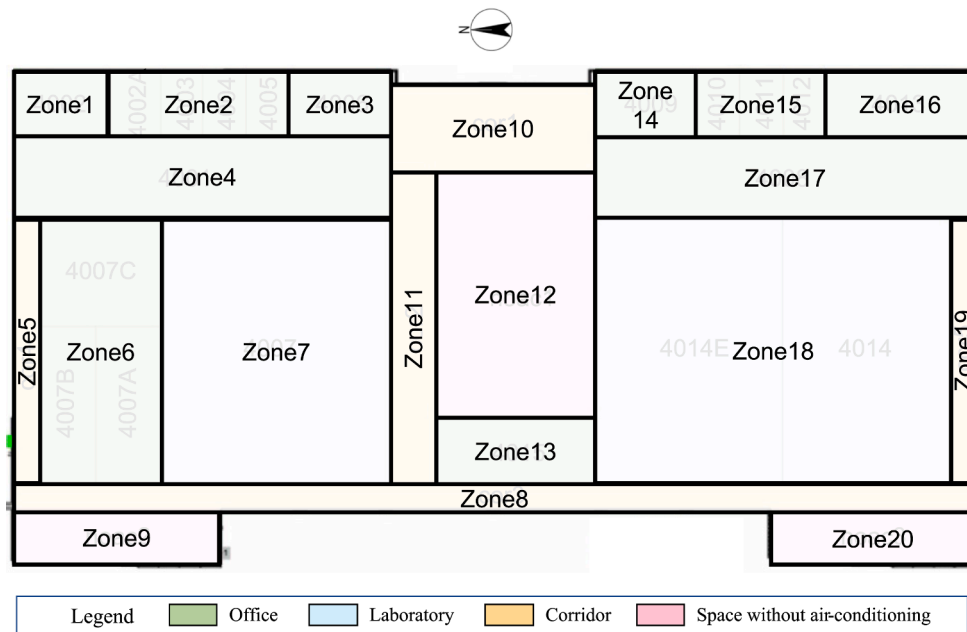
Fig. 12 shows the simulation result of the BEM model generated by the proposed framework. The period from 7 November 2022 to 13 November 2022 was within the academic time of the fall semester and could represent the typical weekly trend of cooling load during common working time in transition seasons. From the simulation results, the building managers could have a clear picture of the energy profile and conduct some preliminary analysis for further decisions. For example, Fig. 12 (a) shows the trend of the hourly cooling loads. The daily peak of cooling load occurred during the daytime, i.e., the working time of the building. There are sharp rises at approximately 7:00–8:00 and sharp drops at approximately 16:00–18:00 on weekdays and Saturday. The cooling load remained at the lowest level during the nighttime and the lowest level of the cooling load was approximately 10 kWh, which might be contributed by the heat gain of the full-time-running equipment and lighting. On Sunday (13 November 2022), the cooling load was the lowest of the week, which might result from less internal heat gain from occupancy. Fig. 12 (b) shows the daily loads of occupancy, lighting and electric equipment, the three aspects the building managers might focus on. The daily contributions of the three internal sensible heat gains were similar in the weekdays and the daily load reduced in the weekend, when the building was less occupied and the electric facilities were less used. The electric equipment took the largest proportion of the daily loads, while occupancy took the smallest. The proportion of the lighting was slightly lower than the electric equipment. Hence, to reduce the cooling load during transition seasons, more attention should be paid to the electric equipment and the lighting.

4.2.3. Framework performance

The goal of the proposed ontology-based framework is to achieve precise and timely automatic BEM. Hence, model precision and modeling time are two main aspects of the framework's performance.



(a) Spaces and expected thermal zoning result (divided by red lines)



(b) Thermal zoning result by reasoning

Fig. 11. Layouts of spaces and thermal zones.

For model precision, the EPW file and IDF file generated by the proposed framework were compared with the files by manual modeling. For the EPW files, the values of each parameter at each timestamp were compared and CV(RMSE) were calculated according to Eq. (1). The CV (RMSE) of the weather parameters was 0, showing the consistency between the EPW files generated by the proposed framework and manual modeling. For the IDF files, firstly the values at each timestamp of each schedule were compared by CV(RMSE). The CV(RMSE) of the schedule was 0, showing the consistency between the automatically generated schedules and manual schedules. Secondly, the vertex coordinates of the building surfaces and fenestration surfaces were compared after the alignment of the coordinate systems by CV(RMSE). The CV(RMSE) of the schedule was 0, showing the consistency of the geometry models generated by the proposed framework and manual modeling. Finally,

the hourly values of the cooling load simulation results were compared by CV(RMSE). The CV(RMSE) of the schedule was 0, showing the consistency of the simulation results between the proposed framework and manual modeling. The results showed the consistency between the automatically generated model and the manual model in terms of weather parameters, schedules, geometric models and simulation results. It can be concluded that the proposed framework can satisfy the precision that the manual model can achieve.

The CV(RMSE) is defined as follows:

$$CV(RMSE)(\%) = \sqrt{\frac{\sum_{i=1}^n (t_{ip} - t_{im})^2}{n-1}} \times \frac{1}{t_{im}} \times 100 \quad (1)$$

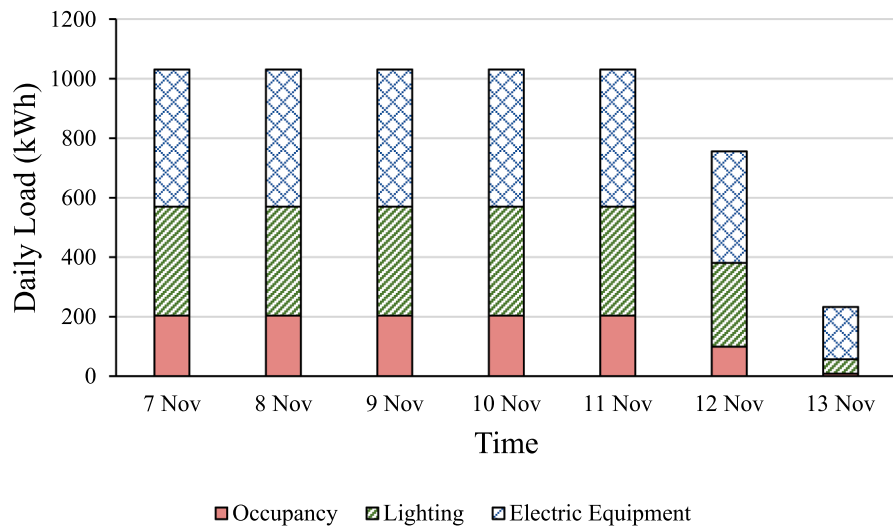
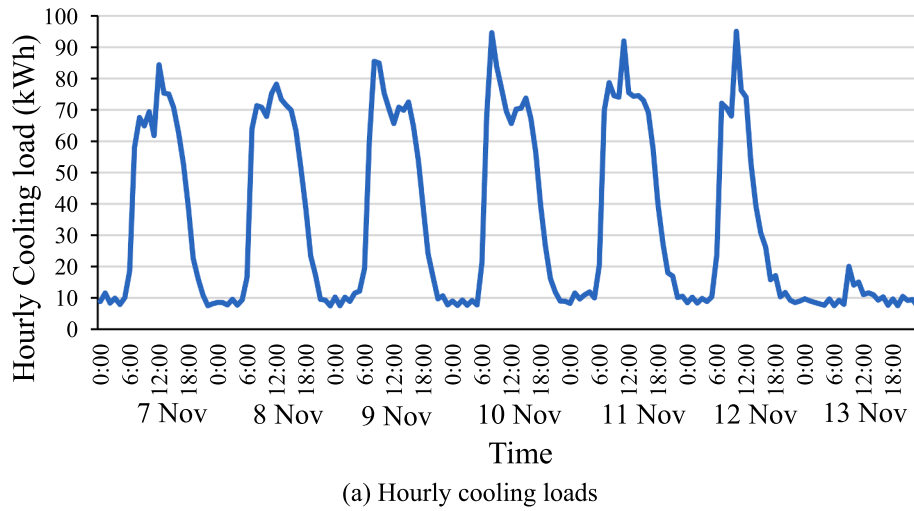


Fig. 12. Building energy simulation results of the BEM model generated by the proposed framework.

where t_{ip} is the i -th value from the proposed framework, t_{im} is i -th value from manual modeling, $\bar{t}_m$ is the arithmetic mean of a sample of n data, and n is the number of the values.

For modeling time, three parts, i.e., ontology modeling, building energy modeling and building energy simulation, should be considered. We adopted manual and semi-automatic methods to develop the ontology model based on information from different sources. It took approximately 26 h to finish the ontology model and 6.5 h to check and refine it. For building energy modeling, the time consumption consisted of three parts, i.e., reasoning, EPW file generation, and IDF file generation. The reasoning was conducted in 0.751 s by Python scripts for the assignment of property `bes:hasExternalWall` and in 0.774 s by Protégé for the assignment of property `bes:sameThermalZone`. The EPW file generation was finished in 0.545 s and the IDF file generation was done in 2.702 s using Python scripts. In totally, 4.772 s was consumed for the building energy modeling. The manual modeling process took approximately 8.5 h for information retrieval from different sources, and EPW and IDF file editing. For building energy simulation, the procedure was conducted in 7.18 s by EnergyPlus [4], and the time was the same between the proposed framework and manual modeling.

Table 6 summarizes the time consumption in each step of manual modeling and the proposed framework in the illustrative example.

Table 6

Time consumption in each step of manual modeling and proposed framework in the case.

	Ontology modeling	Building energy modeling	Building energy simulation
Manual modeling	0	8.5 h	7.18 s
Proposed framework (first time)	32.5 h	4.772 s	7.18 s
Proposed framework (following tasks)	0	4.772 s	7.18 s

Although the proposed framework took more time in total, the time consumption of ontology modeling took the largest proportion. To establish the proposed framework for building energy simulation, the ontology model should be developed for the first time. However, besides small revisions of the model according to some information changes, the ontology can be reused in the following simulation tasks. Consequently, for the following simulation tasks, only the time consumption of building energy modeling and building energy simulation should be considered. The building energy modeling can be achieved by the automatic framework within few seconds. Although some parts of the manual energy simulation model can be reused in the following simulation

tasks, it still needs to retrieve and prepare the dynamic data, e.g., weather and schedules. The results of thermal zoning will change if space types or schedules of some spaces alter, and hence the thermal zones are still needed to be modified in the following simulation tasks. For the following tasks, the modeling time can be reduced by over 99% ($|4.772 \text{ s} - 8.5 \text{ h}| / 8.5 \text{ h} \times 100\%$) by the proposed framework compared with the manual modeling.

To summary, the proposed framework can keep the same precision as the manual modeling while substantially reducing the BEM time consumption by over 99% once the ontology model is established.

4.3. Discussion

4.3.1. Data integration

To evaluate completeness of the data integration by the proposed framework by comparing with the current IFC- or gbXML-based methods, we followed Gao et al.'s BIM-BEM process classification [6] and inspected the degree of completeness in this study as shown in Table 7. The data of geometry, construction and materials, and building or space types are integrated from BIM. They are covered by the proposed framework. Thermal zoning is achieved by reasoning of the ontology model. For space load, sensible heat gain including occupancy, lighting and electric equipment, and their schedules are integrated in this study, but there is a lack of latent heat gain, ventilation load and infiltration load. For HVAC system, data integration of ideal HVAC systems and their schedules are achieved in this study and the complex HVAC systems need to be further considered. In summary, the proposed framework could complete the first four steps for BEM while partly achieving the fifth and sixth steps. According to Gao et al.'s review [6], the IFC-based studies can achieve Step 2, and gbXML-based studies can achieve Step 4 on average. The proposed framework could outperform the average levels of the current IFC- or gbXML-based methods in terms of data integration.

4.3.2. Implications

Fig. 13 shows the position of this study in the field of automatic BEM. Current studies focused on BIM-based BEM, i.e., to improve BIM-based schemas, e.g., IFC and gbXML, to enhance the interoperability between BIM and BEM. According to ASHRAE Research Project RP-1468 [36], BIM should not be considered as a complete database for the information necessary for building energy simulation tools since it does not have information such as complete material thermal properties, weather data, scheduling information, and internal loads. From the perspective of this study, the goal is not to develop a full BIM to achieve automatic BEM. Instead, each information source has its focus, and BIM could serve as one of the information sources and provides mainly the information related to building geometries and construction properties. Other information, e.g., occupancy and complex schedules can be

provided by other sources, e.g., BMS and related databases.

Ontology, the tool-independent and vendor-free model, can serve as the bridge among different information domains. In this study, four main information types for BEM, i.e., weather, building, internal heat gain and HVAC system, were transferred to the ontology model. Knowledge of BEM could be stored in the proposed ontology model in structured data schema. Besides, ontology belongs to open formats. Current ontology models (e.g., BOT and Brick Schema in this study) can be reused and extension can be supplemented to the proposed ontology model while necessary in the future (e.g., latent heat gain, ventilation load, etc. which the current study lacks as mentioned in Section 4.3.1). The dynamic data conversion issue could be achieved by the ontology framework. The dynamic data are stored at their original addresses and in their native format, and dynamic data conversion algorithms could be implemented to retrieve the data based on the link of the ontology model. Results shows that the dynamic data converted by the proposed framework can keep consistency with manual modeling. In addition, this study takes one aggregation rule with four criteria for thermal zoning for example and results shows that the cross-domain and rule-based thermal zoning could be achieved based on reasoning of the ontology model.

It should be clarified that the automatic translation from various domains, e.g., BIM and BMS, to the ontology model (the arrow in the dash line in Fig. 13) is not within the scope of this study. The translation to ontology can be implemented by some existing methods. For example, the ifcOWL and its tools [50] could facilitate the translation from BIM to ontology.

4.3.3. Limitations and future work

Some limitations are identified in this study and more improvements should be considered in the future studies.

(1) Aggregation and separation are two methods for thermal zoning. The aggregation method is commonly used in buildings consisting of multiple spaces with similar attributes, which the proposed framework can be applied to. For particular spaces with non-uniform attributes (e.g., temperature) distribution, e.g., atriums and large open offices, the separation method should be studied in the future.

(2) Although the main parts of BEM were included in this study, other important components, e.g., shading, infiltration and ventilation for occupants, were not considered. Further extensions on the ontology model are needed to include those components.

(3) Due to the security and privacy concern by the campus management office, details of HVAC specifications and data are not allowed to be revealed to the public and only VAV setpoints and schedules are available in the illustrative example. Hence, this study focused on the thermal load simulation in which the zone temperatures and schedules were mainly concerned. Future work should include a more comprehensive analysis of HVAC specifications and data.

(4) Only manual modeling method was compared in terms of framework performance. IFC- or gbXML-based methods which took thermal zoning and dynamic data conversion (e.g., [37,73–77]) may be included. However, to the best knowledge of the authors, there are no open tools or scripts that support these IFC- or gbXML-based methods, which makes the reproduction difficult. Future work should consider the comparison of IFC- or gbXML-based methods when there are more open-source tools or scripts.

(5) For dynamic data, only weather data and schedules of equipment and occupancy were considered in this study. In the future, more data could be included. For example, historical indoor conditions should be included when BEM is conducted for calibration.

5. Conclusions

This study proposed an ontology-based automatic framework aimed at generating BEM model in an automatic way by integrating multiple data sources. Firstly, an ontology model was proposed to integrate data

Table 7
Degree of completeness for data integration.

Step	Information source	Completion	Lack
Geometry	BIM	✓	–
Construction and materials	BIM	✓	–
Building or space types	BIM	✓	–
Thermal zones	By reasoning	✓	–
Space loads	BIM, BMS, and sensors	Partly: Sensible heat gain including occupancy, lighting and electric equipment; and schedule	Latent heat gain, ventilation load, and infiltration load
HVAC system and its components	BMS	Partly: Ideal HVAC system; and schedule	Complex HVAC system

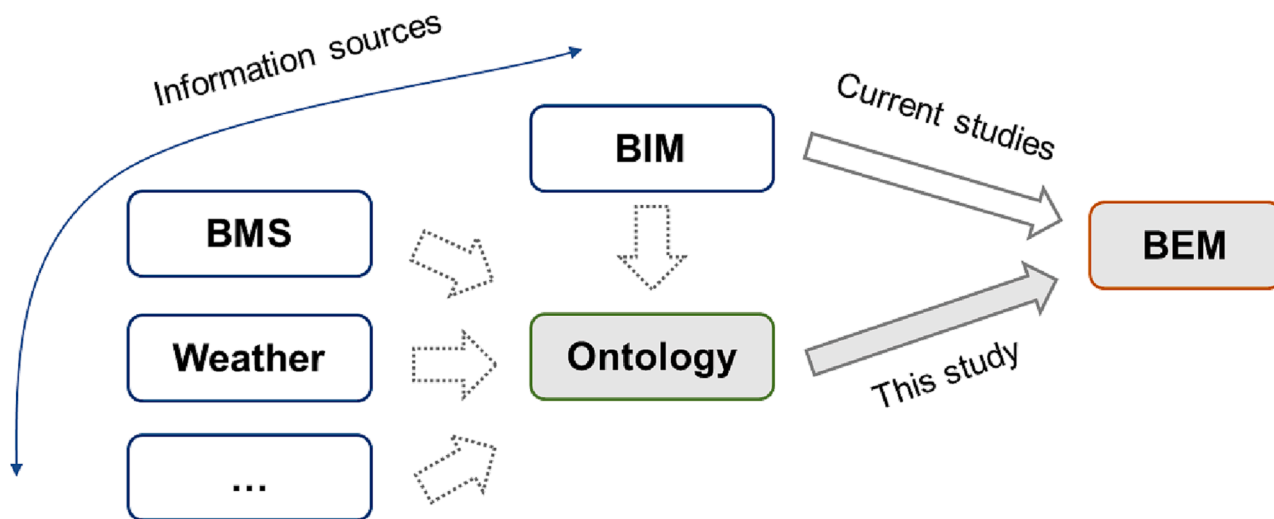


Fig. 13. The position of this study in the field of automatic BEM.

from four sources that are critical for BEM including weather, building, internal heat gain, and HVAC system. The ontology model was design based on three principles, i.e., (1) reuse of existing ontology models (Brick Schema and BOT) with necessary extension, (2) domain-specific design, and (3) linked with dynamic data. Secondly, a cross domain and rule-based method for thermal zoning was proposed based on reasoning of the proposed ontology model. Thirdly, a method of model translation from the ontology model to the BEM model was proposed based on instance-based mapping, dynamic data conversion and template-based configuration file generation. We demonstrated how this framework can be used to automatically generate a BEM model for one floor of a campus building for cooling load simulation using one-week meteorological data of a transition season in Hong Kong. We summarize the major conclusions of this study as follows.

- The proposed ontology model can integrate data from multiple information sources of weather, building, internal heat gain and HVAC system which are key inputs for BEM.
- The proposed framework can achieve automatic cross-domain and rule-based thermal zoning using ontology-based reasoning based on

criteria of internal heat gain, external heat gain and HVAC control. The reasoning results comply well with the standards or guidelines [39–41], and the proposed automatic framework allows a fast and accurate thermal zoning process.

- Dynamic data (including weather and schedules) can be automatically converted to the BEM model by the proposed framework, and the data keep consistency with manual modeling.
- The proposed framework can satisfy the modeling precision that the manual model can achieve, and substantially reduce the modeling time by more than 99% comparing with manual modeling.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Appendix A. Graph of the ontology model

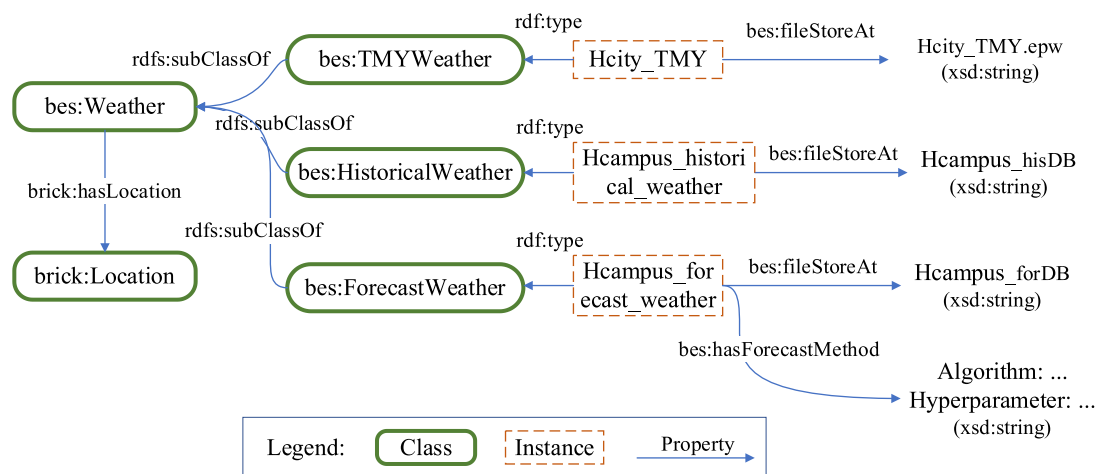


Fig. A.1. Weather ontology model.

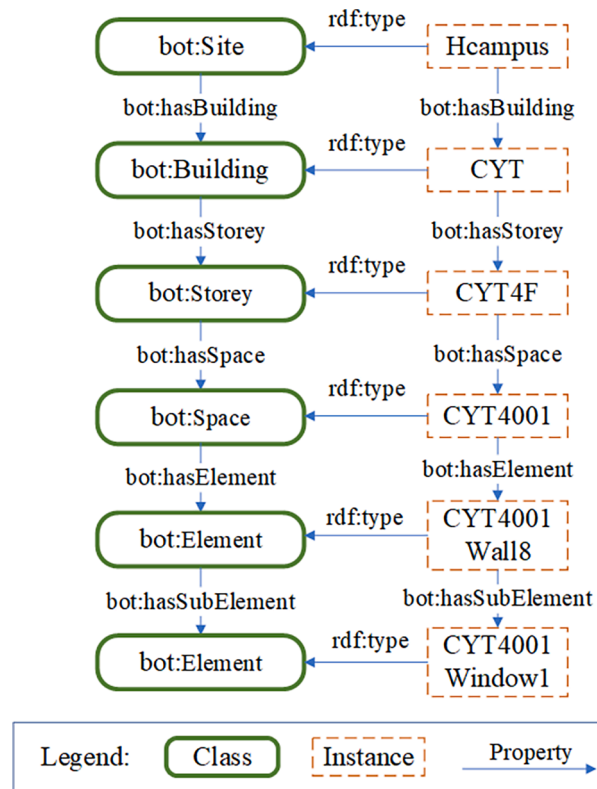


Fig. A.2. Ontology model of spatial units based on BOT.

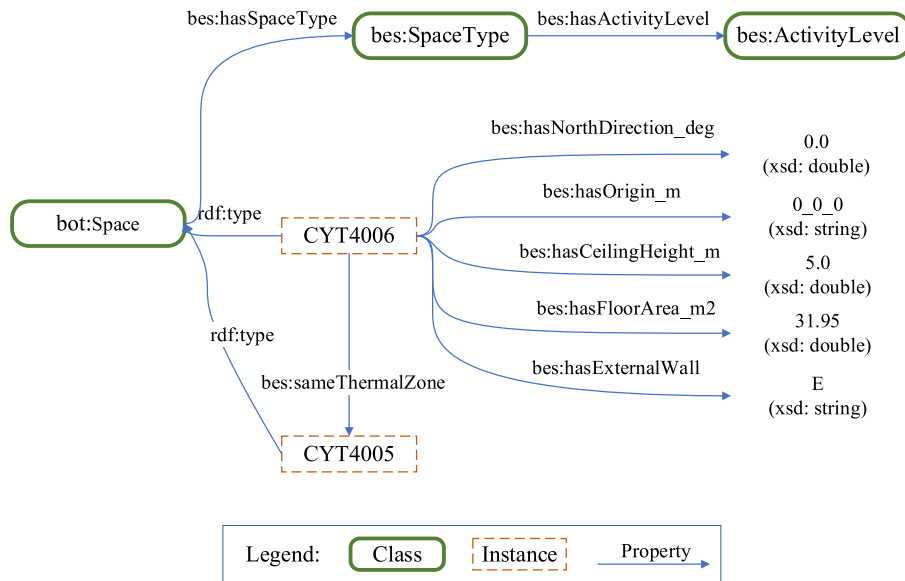
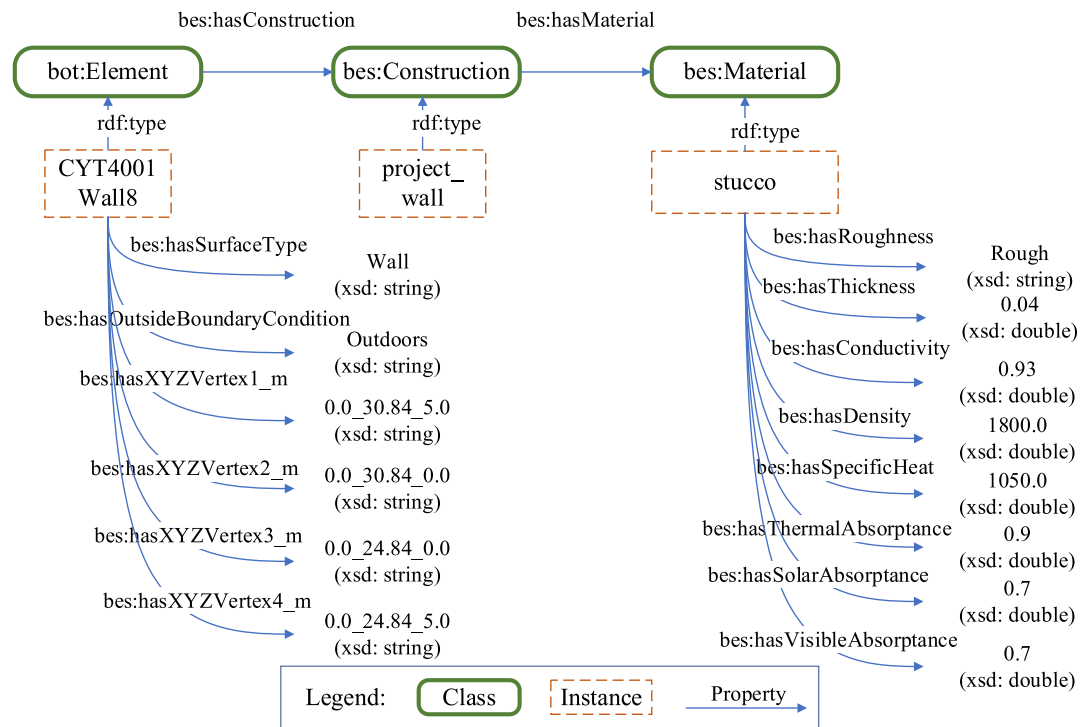
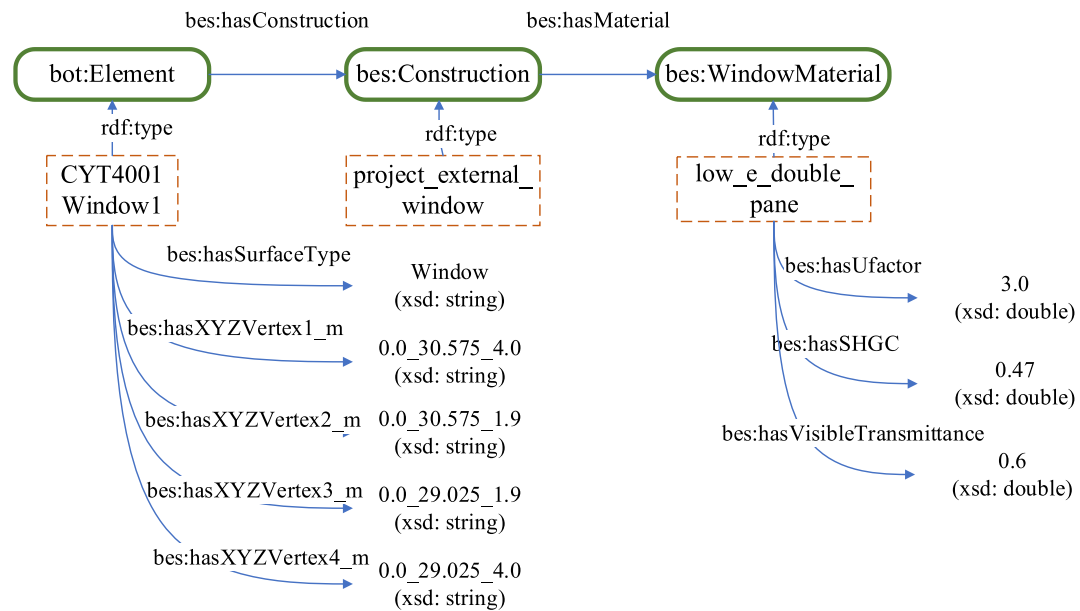


Fig. A.3. Extension of bot:Space for BEM.

Fig. A.4. Extension of `bot:Element` for building surfaces.Fig. A.5. Extension of `bot:Element` for fenestration surfaces.

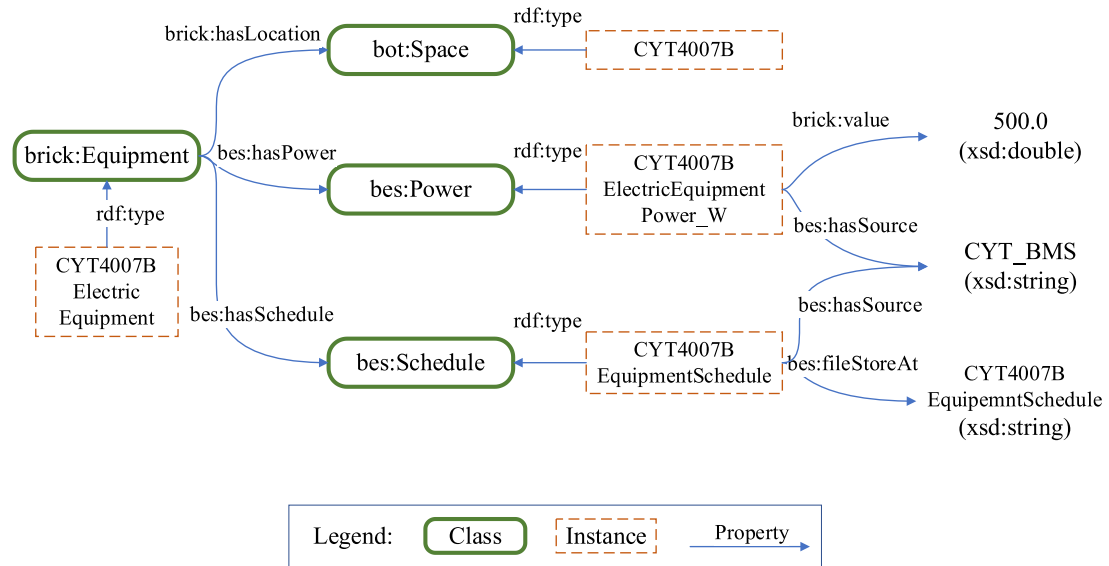


Fig. A.6. Ontology model of equipment.

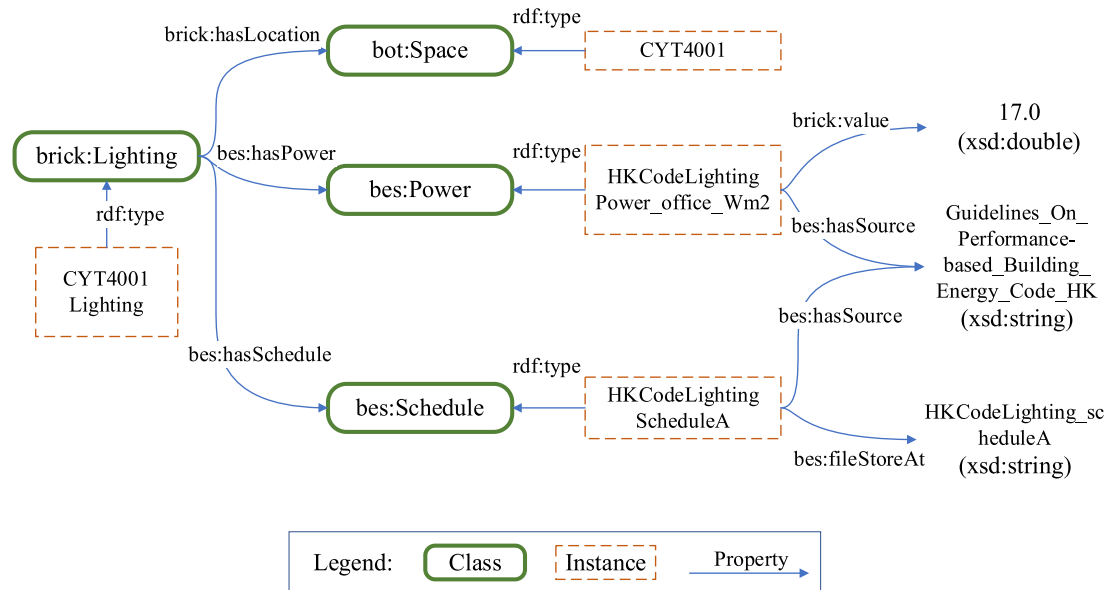


Fig. A.7. Ontology model of lighting.

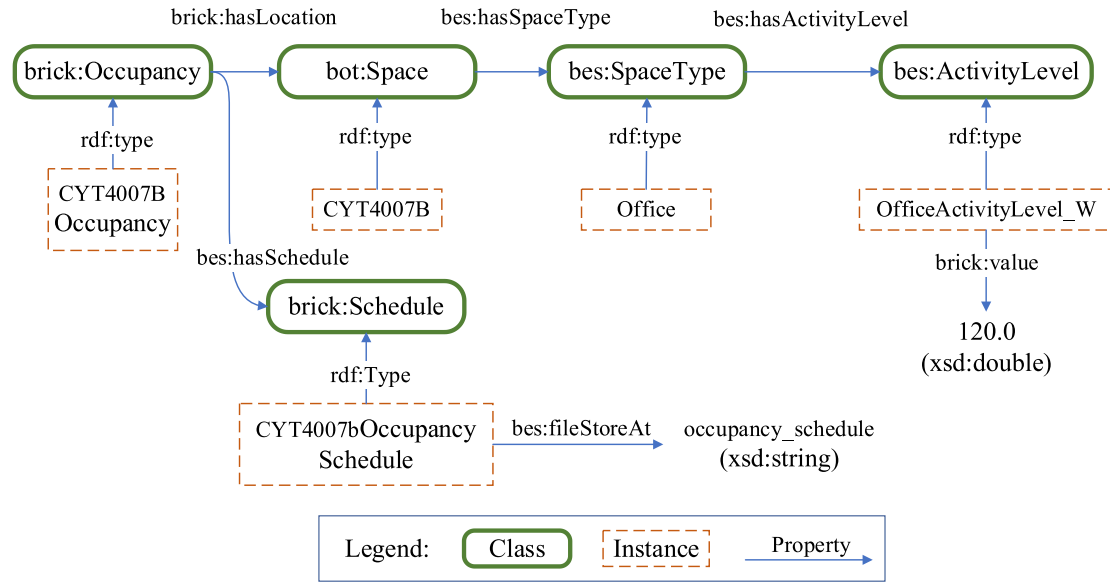


Fig. A.8. Ontology model of occupancy.

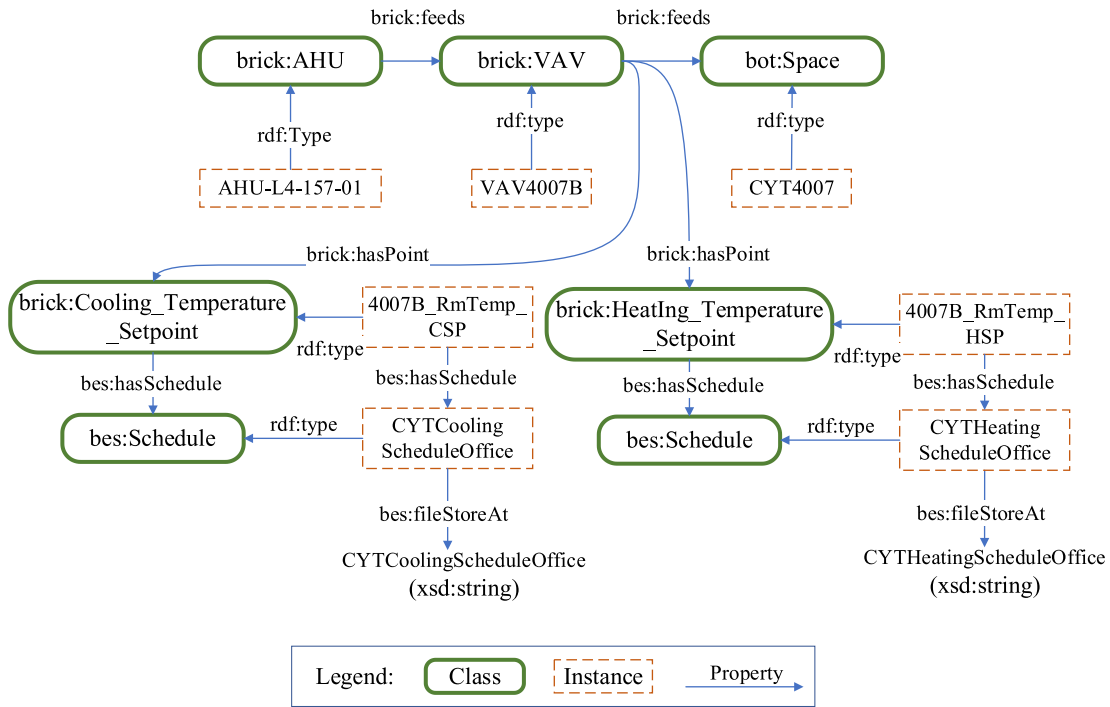


Fig. A.9. Ontology model of HVAC system.

Appendix B. Entity mapping and property mapping between the proposed ontology model and BEM model for thermal load simulation

Table B1

Object mapping between the proposed ontology model and the building energy simulation model.

IDF object	Ontology object
Material	bes:Material
Roughness	bes:hasRoughness
Thickness	bes:hasThickness
Conductivity	bes:hasConductivity
Density	bes:hasDensity
Specific Heat	bes:hasSpecificHeat
Thermal Absorptance	bes:hasThermalAbsorptance

(continued on next page)

Table B1 (continued)

IDF object	Ontology object
Solar Absorptance	bes:hasSolarAbsorptance
Visible Absorptance	bes:hasVisibleAbsorptance
WindowMaterial:SimpleGlazingSystem	bes:WindowMaterial
U-Factor	bes:hasUfactor
Solar Heat Gain Coefficient	bes:hasSHGC
Visible Transmittance	bes:hasVisibleTransmittance
Construction	bes:Construction
Outside Layer	bes:hasMaterial (first layer)
Layer 2	bes:hasMaterial (second layer)
...	...
Space	bot:Space
Zone Name	Generated by thermal zoning algorithm (Section 3.2)
Floor Area	Default (autocalculate)
Space Type	bes:hasSpaceType
Zone	Generated by thermal zoning algorithm (Section 3.2)
Ceiling Height	Default (autocalculate)
Volume	Default (autocalculate)
Floor Area	Default (autocalculate)
BuildingSurface:Detailed	bot:Elenment
Surface Type	bes:hasSurfaceType
Construction Name	bes:hasConstruction
Zone Name	Generated by thermal zoning algorithm (Section 3.2)
Space Name	bot:hasElenment
Outside Boundary Conditon	bes:hasOutsideBoundaryCondition
Sun Exposure	Generated based on Outside Boundary Conditon
Wind Exposure	Generated based on Outside Boundary Conditon
Vertex 1 X-coordinate	bes:hasXYZVertex1_m
Vertex 1 Y-coordinate	bes:hasXYZVertex1_m
Vertex 1 Z-coordinate	bes:hasXYZVertex1_m
Vertex 2 X-coordinate	bes:hasXYZVertex2_m
Vertex 2 Y-coordinate	bes:hasXYZVertex2_m
Vertex 2 Z-coordinate	bes:hasXYZVertex2_m
Vertex 3 X-coordinate	bes:hasXYZVertex3_m
Vertex 3 Y-coordinate	bes:hasXYZVertex3_m
Vertex 3 Z-coordinate	bes:hasXYZVertex3_m
Vertex 4 X-coordinate	bes:hasXYZVertex4_m
Vertex 4 Y-coordinate	bes:hasXYZVertex4_m
Vertex 4 Z-coordinate	bes:hasXYZVertex4_m
FenestrationSurface:Detailed	bot:Element
Surface Type	bes:hasSurfaceType
Construction Name	bes:hasConstruction
Building Surface Name	bot:hasSubElenment
Vertex 1 X-coordinate	bes:hasXYZVertex1_m
Vertex 1 Y-coordinate	bes:hasXYZVertex1_m
Vertex 1 Z-coordinate	bes:hasXYZVertex1_m
Vertex 2 X-coordinate	bes:hasXYZVertex2_m
Vertex 2 Y-coordinate	bes:hasXYZVertex2_m
Vertex 2 Z-coordinate	bes:hasXYZVertex2_m
Vertex 3 X-coordinate	bes:hasXYZVertex3_m
Vertex 3 Y-coordinate	bes:hasXYZVertex3_m
Vertex 3 Z-coordinate	bes:hasXYZVertex3_m
Vertex 4 X-coordinate	bes:hasXYZVertex4_m
Vertex 4 Y-coordinate	bes:hasXYZVertex4_m
Vertex 4 Z-coordinate	bes:hasXYZVertex4_m
People	brick:Occupancy
Zone or Zonelist or Space or Spacelist Name	brick:hasLocation
Number of People Schedule Name	bes:hasSchedule
Number of People Calculation method	“People/Area”
People per Floor Area	Calculated or retrieved from guidelines or standards
Activity Level Schedule Name	bes:hasActivityLevel
Lights	brick:Lighting_Equipment
Zone or Zonelist or Space or Spacelist Name	brick:hasLocation
Schedule Name	bes:hasSchedule
Design Level Calculation method	“Watts/Area”
Watts per Zone Floor Area	Calculated or retrieved from guidelines or standards
ElectricEquipment	brick:Electrical_Equipment
Zone or Zonelist or Space or Spacelist Name	brick:hasLocation
Schedule Name	bes:hasSchedule
Design Level Calculation method	“Watts/Area”
Watts per Zone Floor Area	Calculated or retrieved from guidelines or standards
ThermostatSetpoint:DualSetpoint	Each zone is assigned with a setpoint
Heating Setpoint Temperature Schedule Name	brick:Cooling_Temperature_Setpoint
Cooling Setpoint Temperature Schedule Name	brick:HeatIng_Temperature_Setpoint

Appendix C. Scripts for material information translation

Table C1

Scripts for material information translation.

Scripts: Material information translation	
1	#file preparing
2	ontology_model = './ontology/energy_simulation.owl'
3	idf_template = 'template.idf'
4	idd_file = 'V9-6-0-Energy +.idd'
5	save_file = 'CYT4F_bes.idf'
6	
7	#OWL preparing
8	g = Graph()
9	g.parse(ontology_model, format='turtle')
10	
11	#IDF preparing
12	iddfile = idd_file
13	fname = idf_template
14	IDF.setiddname(iddfile)
15	idf = IDF(fname)
16	
17	query_material = '''
18	SELECT?material?Roughness?Thickness?Conductivity?Density?SpecificHeat?ThermalAbsorptance?SolarAbsorptance?VisibleAbsorptance
19	WHERE {
20	?material rdf:type bes:Material.
21	OPTIONAL {?material bes:hasRoughness?Roughness. }
22	OPTIONAL {?material bes:hasThickness?Thickness. }
23	OPTIONAL {?material bes:hasConductivity?Conductivity. }
24	OPTIONAL {?material bes:hasDensity?Density. }
25	OPTIONAL {?material bes:hasSpecificHeat?SpecificHeat. }
26	OPTIONAL {?material bes:hasThermalAbsorptance?ThermalAbsorptance. }
27	OPTIONAL {?material bes:hasSolarAbsorptance?SolarAbsorptance. }
28	OPTIONAL {?material bes:hasVisibleAbsorptance?VisibleAbsorptance. }
29	}
30	'''
31	
32	qres_material = g.query(query_material)
33	
34	for row in qres_material:
35	material_name_list = ['material_name', 'Roughness', 'Thickness', 'Conductivity', 'Density', 'SpecificHeat', 'ThermalAbsorptance', 'SolarAbsorptance', 'VisibleAbsorptance']
36	material_info_list = []
37	for i in range(len(row)):
38	if str(type(row[i])) == "<class 'rdflib.term.URIRef'>":
39	material_info_list.append(str(row[i]).split('#')[-1])
40	else:
41	#elif str(type(row[i])) == "<class 'rdflib.term.Literal'>":
42	material_info_list.append(str(row[i]))
43	
44	material_dict = {}
45	for i in range(len(row)):
46	material_dict[material_name_list[i]] = material_info_list[i]
47	if material_info_list[i] == 'None':
48	print('%s lacks information of %s' % (material_info_list[0], material_name_list[i]))
49	else:
50	pass
51	
52	new_Material = idf.newidfobject('Material',
53	Name = material_dict.get('material_name'),
54	Roughness = material_dict.get('Roughness'),
55	Thickness = material_dict.get('Thickness'),
56	Conductivity = material_dict.get('Conductivity'),
57	Density = material_dict.get('Density'),
58	Specific_Heat = material_dict.get('SpecificHeat'),
59	Thermal_Absorptance = material_dict.get('ThermalAbsorptance'),
60	Solar_Absorptance = material_dict.get('SolarAbsorptance'),
61	Visible_Absorptance = material_dict.get('VisibleAbsorptance')
62	)
63	
64	idf.saveas(save_file)

References

- [1] IEA, Buildings, IEA, Paris, 2022. <https://www.iea.org/reports/buildings>.
- [2] ASHRAE, Handbook of Fundamentals: Energy Estimating and Modeling Methods; SI Edition, 2009.
- [3] K. Amasyali, N.M. El-Gohary, A review of data-driven building energy consumption prediction studies, *Renew. Sustain. Energy Rev.* 81 (2018) 1192–1205, <https://doi.org/10.1016/j.rser.2017.04.095>.
- [4] U.S. Department of Energy's (DOE) Building Technologies Office (BTO), National Renewable Energy Laboratory (NREL), EnergyPlus, (2023). <https://energyplus.net/> (accessed May 18, 2023).

- [5] James J. Hirsch, eQuest, (2018). <https://www.doe2.com/equest/> (accessed March 8, 2023).
- [6] H. Gao, C. Koch, Y. Wu, Building information modelling based building energy modelling: a review, *Appl. Energy* 238 (2019) 320–343, <https://doi.org/10.1016/j.apenergy.2019.01.032>.
- [7] H. Zhao, F. Magoulès, A review on the prediction of building energy consumption, *Renew. Sustain. Energy Rev.* 16 (2012) 3586–3592, <https://doi.org/10.1016/j.rser.2012.02.049>.
- [8] M. Weise, M. Poveda Villalón, R. García Castro, J. Euzenat, L.M. Priego, B. Fies, A. Cavallaro, J. Peters-Anders, K. Zoi Tsagkari, Ontologies and datasets for energy management system interoperability, *Ready4SmartCities*, 2015. <https://inria.hal.science/hal-01247606>.
- [9] P. Pauwels, S. Zhang, Y.-C. Lee, Semantic web technologies in AEC industry: a literature overview, *Autom. Constr.* 73 (2017) 145–165, <https://doi.org/10.1016/j.autcon.2016.10.003>.
- [10] Y.-Y. Zhang, Z.-Z. Hu, J.-R. Lin, J.-P. Zhang, Linking data model and formula to automate KPI calculation for building performance benchmarking, *Energy Rep.* 7 (2021) 1326–1337, <https://doi.org/10.1016/j.egyr.2021.02.044>.
- [11] E. Corry, P. Pauwels, S. Hu, M. Keane, J. O'Donnell, A performance assessment ontology for the environmental and energy management of buildings, *Autom. Constr.* 57 (2015) 249–259, <https://doi.org/10.1016/j.autcon.2015.05.002>.
- [12] S. Hu, J. Wang, C. Hoare, Y. Li, P. Pauwels, J. O'Donnell, Building energy performance assessment using linked data and cross-domain semantic reasoning, *Autom. Constr.* 124 (2021), 103580, <https://doi.org/10.1016/j.autcon.2021.103580>.
- [13] J. Han, Y.-K. Jeong, I. Lee, A Rule-Based Ontology Reasoning System for Context-Aware Building Energy Management, in: 2015 IEEE International Conference on Computer and Information Technology; Ubiquitous Computing and Communications; Dependable, Autonomic and Secure Computing; Pervasive Intelligence and Computing, IEEE, Liverpool, United Kingdom, 2015: pp. 2134–2142. doi: 10.1109/CIT/IUCC/DASC/PICOM.2015.317.
- [14] H.E. Degha, F.Z. Laallam, B. Said, Intelligent context-awareness system for energy efficiency in smart building based on ontology, *Sustainable Comput. Inf. Syst.* 21 (2019) 212–233, <https://doi.org/10.1016/j.suscom.2019.01.013>.
- [15] P. Pauwels, E. Corry, J. O'Donnell, Representing SimModel in the Web Ontology Language, in: *Computing in Civil and Building Engineering* (2014), American Society of Civil Engineers, Orlando, Florida, United States, 2014: pp. 2271–2278. doi: 10.1061/9780784413616.282.
- [16] A. Farzaneh, D. Monfet, D. Forgues, Review of using Building Information Modeling for building energy modeling during the design process, *J. Build. Eng.* 23 (2019) 127–135, <https://doi.org/10.1016/j.job.2019.01.029>.
- [17] E. Kamel, A.M. Memari, Review of BIM's application in energy simulation: Tools, issues, and solutions, *Autom. Constr.* 97 (2019) 164–180, <https://doi.org/10.1016/j.autcon.2018.11.008>.
- [18] buildingSMART, Industry Foundation Classes (IFC), (2019). <https://www.buildingsmart.org/standards/bsi-standards/industry-foundation-classes/> (accessed May 18, 2023).
- [19] Green Building XML (gbXML) Schema, Inc., Green Building XML (gbXML), (2023). <https://www.gbxml.org/> (accessed March 8, 2023).
- [20] G.N. Lilis, G.I. Giannakis, D.V. Rovas, Automatic generation of second-level space boundary topology from IFC geometry inputs, *Autom. Constr.* 76 (2017) 108–124.
- [21] K.-U. Ahn, Y.-J. Kim, C.-S. Park, I. Kim, K. Lee, BIM interface for full vs. semi-automated building energy simulation, *Energy. Build.* 68 (2014) 671–678.
- [22] A. Karola, H. Lahtela, R. Hänninen, R. Hitchcock, Q. Chen, S. Dajka, K. Hagström, B3Pro COM-Server—interoperability between software tools using industrial foundation classes, *Energy. Build.* 34 (9) (2002) 901–907.
- [23] R. See, P. Haves, P. Sreekanthan, J. O'Donnell, M. Basarkar, K. Settlemeyre, Development of a user interface for the EnergyPlus whole building energy simulation program, in: *Proceedings of the 12th International IBPSA Conference*, Sydney, Australia, 2011.
- [24] J. O'Donnell, R. See, C. Rose, T. Maile, V. Bazjanac, P. Haves, SimModel: A domain data model for whole building energy simulation, in: *IBPSA Building Simulation 2011*, 2011.
- [25] G. Giannakis, G. Lilis, M. Garcia, G. Kontes, C. Valmaseda, D. Rovas, A methodology to automatically generate geometry inputs for energy performance simulation from IFC BIM models, in: *14th International Conference of IBPSA-Building Simulation 2015, BS 2015, Conference Proceedings, IBPSA, Hyderabad, India*, 2015: pp. 504–511.
- [26] A. Cemesova, C.J. Hopfe, R.S. Mcleod, PassivBIM: Enhancing interoperability between BIM and low energy design software, *Autom. Constr.* 57 (2015) 17–32, <https://doi.org/10.1016/j.autcon.2015.04.014>.
- [27] A. Cemesova, C. Hopfe, Y. Rezgui, An approach to facilitating data exchange between BIM environments and a low energy design tool, in: *Proceedings of the BS2013, 13th Conference of International Building Performance Simulation Association*, Chambéry, France, 2013: pp. 25–30.
- [28] A. Cemesova, Enhancing BIM-based data transfer to support the design of low energy buildings, PhD Thesis, Cardiff University, 2013.
- [29] I. Kim, J. Kim, J. Seo, Development of an IFC-based IDF converter for supporting energy performance assessment in the early design phase, *J. Asian Architecture Build. Eng.* 11 (2) (2012) 313–320.
- [30] J.C. Cheng, M. Das, A BIM-based web service framework for green building energy simulation and code checking, *J. Information Technol. Constr. (ITcon)* 19 (2014) 150–168.
- [31] V. Bazjanac, T. Maile, IFC HVAC interface to EnergyPlus-A case of expanded interoperability for energy simulation, Lawrence Berkeley National Lab.(LBNL), Berkeley, CA (United States), 2004.
- [32] B. O'Sullivan, M. Keane, Specification of an IFC based intelligent graphical user interface to support building energy simulation, in: *Proceedings of the Ninth International IBPSA Conference*, Citeseer, Montréal, Canada, 2005: pp. 15–18.
- [33] S.V. Pinheiro, R. Wimmer, T. Maile, J. O'Donnell, others, Model view definition for advanced building energy performance simulation, in: *CESBP/BausIM 2016: CESBP Central European Symposium on Building Physics*, Technische Universität Dresden, Germany, 14–16 September 2016, 2016.
- [34] F. Jalaei, A. Jrade, Integrating building information modeling (BIM) and energy analysis tools with green building certification system to conceptually design sustainable buildings, *J. Inf. Technol. Constr.* 19 (2014) 494–519.
- [35] J. Han, Y.-K. Jeong, I. Lee, Efficient building energy management system based on ontology, inference rules, and simulation, in: *Proceedings of the 2011 International Conference on Intelligent Building and Management*, Singapore, 2011.
- [36] M. Clayton, J. Haberl, W. Yan, S. Kota, F. Farias, W.S. Jeong, J.B. Kim, J.L. Bermudez Alcocer, Development of a reference building information model (BIM) for thermal model compliance testing, *ASHRAE*, 2013.
- [37] E. Kamel, A.M. Memari, Automated Building Energy Modeling and Assessment Tool (ABEMAT), *Energy* 147 (2018) 15–24, <https://doi.org/10.1016/j.energy.2018.01.023>.
- [38] M. Shin, J.S. Haberl, A procedure for automating thermal zoning for building energy simulation, *J. Build. Eng.* 46 (2022), 103780, <https://doi.org/10.1016/j.job.2021.103780>.
- [39] S. Edition, D. Erbe, M. Lane, S. Anderson, P. Baselici, S. Hanson, R. Heinisch, J. Humble, S. Taylor, R. Kurtz, Energy standard for buildings except low-rise residential buildings, in: *ASHRAE*, Atlanta, GA, USA, Tech. Rep. IP Edition, 2010.
- [40] IBPSA, Thermal Zoning Determination, IBPSA-USA, 2012.
- [41] C.I. of B.S. Engineers (CIBSE), Application Manual AM11: Building Energy and Environmental Modelling, (1998).
- [42] M. Shin, J.S. Haberl, Thermal zoning for building HVAC design and energy simulation: a literature review, *Energy. Build.* 203 (2019), 109429, <https://doi.org/10.1016/j.enbuild.2019.109429>.
- [43] C.M. Rose, V. Bazjanac, An algorithm to generate space boundaries for building energy simulation, *Eng. Comput.* 31 (2) (2015) 271–280.
- [44] D. Ladenhauf, K. Battisti, R. Berndt, E. Eggeling, D.W. Fellner, M. Gratzl-Michlmair, T. Ullrich, Computational geometry in the context of building information modeling, *Energy. Build.* 115 (2016) 78–84.
- [45] T. El-Diraby, T. Krijnen, M. Papagelis, BIM-based collaborative design and socio-technical analytics of green buildings, *Autom. Constr.* 82 (2017) 59–74.
- [46] H. Ying, S. Lee, Generating second-level space boundaries from large-scale IFC-compliant building information models using multiple geometry representations, *Autom. Constr.* 126 (2021), 103659, <https://doi.org/10.1016/j.autcon.2021.103659>.
- [47] X. Chen, S. Jia, Y. Xiang, A review: knowledge reasoning over knowledge graph, *Expert Syst. Appl.* 141 (2020), 112948, <https://doi.org/10.1016/j.eswa.2019.112948>.
- [48] B. Balaji, A. Bhattacharya, G. Fierro, J. Gao, J. Gluck, D. Hong, A. Johansen, J. Koh, J. Ploennings, Y. Agarwal, M. Bergés, D. Culler, R.K. Gupta, M.B. Kjærgaard, M. Srivastava, K. Whitehouse, Brick: Metadata schema for portable smart building applications, *Appl. Energy* 226 (2018) 1273–1292, <https://doi.org/10.1016/j.apenergy.2018.02.091>.
- [49] M.H. Rasmussen, M. Lefrançois, G.F. Schneider, P. Pauwels, K. Janowicz, BOT: The building topology ontology of the W3C linked building data group, *Semantic Web.* 12 (1) (2010) 143–161.
- [50] P. Pauwels, W. Terkaj, EXPRESS to OWL for construction industry: Towards a recommendable and usable ifcOWL ontology, *Autom. Constr.* 63 (2016) 100–133, <https://doi.org/10.1016/j.autcon.2015.12.003>.
- [51] Brick Consortium, Inc., Brick Ontology, (2023). <https://brickschema.org/ontology/> (accessed May 18, 2023).
- [52] EMSD, Guidelines on performance-based building energy code, (2007).
- [53] J.F. Kreider, *Handbook of Heating, Ventilation, and Air Conditioning*, CRC Press, 2000.
- [54] I. Horrocks, P.F. Patel-Schneider, H. Boley, S. Tabet, B. Grosz, M. Dean, others, SWRL: A semantic web rule language combining OWL and RuleML, *W3C Member Submission*. 21 (2004) 1–31.
- [55] H. Li, J. Zhang, S. Chang, A. Sparkling, BIM-based object mapping using invariant signatures of AEC objects, *Autom. Constr.* 145 (2023), 104616, <https://doi.org/10.1016/j.autcon.2022.104616>.
- [56] M. Herrlich, H. Holle, R. Malaka, Integration of cityGML and collada for high-quality geographic data visualization on the PC and Xbox 360, in: *International Conference on Entertainment Computing*, Springer, Seoul, Korea, 2010, pp. 270–277.
- [57] R.R. Lipman, Details of the mappings between the CIS/2 and IFC product data models for structural steel, *J. Information Technol. Constr.* 14 (2009) 1–13.
- [58] Y. Deng, J.C.P. Cheng, C. Anumba, Mapping between BIM and 3D GIS in different levels of detail using schema mediation and instance comparison, *Autom. Constr.* 67 (2016) 1–21, <https://doi.org/10.1016/j.autcon.2016.03.006>.
- [59] R. de Laat, L. van Berlo, Integration of BIM and GIS: The Development of the CityGML GeoBIM Extension, in: T.H. Kolbe, G. König, C. Nagel (Eds.), *Advances in 3D Geo-Information Sciences*, Springer Berlin Heidelberg, Berlin, Heidelberg, 2011: pp. 211–225. doi: 10.1007/978-3-642-12670-3_13.
- [60] M. El-Mekawy, A. Östman, K. Shahzad, Towards Interoperating CityGML and IFC Building Models: A Unified Model Based Approach, in: T.H. Kolbe, G. König, C. Nagel (Eds.), *Advances in 3D Geo-Information Sciences*, Springer Berlin Heidelberg, Berlin, Heidelberg, 2011: pp. 73–93. doi: 10.1007/978-3-642-12670-3_5.

- [61] U. Isikdag, S. Zlatanova, Towards Defining a Framework for Automatic Generation of Buildings in CityGML Using Building Information Models, in: J. Lee, S. Zlatanova (Eds.), 3D Geo-Information Sciences, Springer Berlin Heidelberg, Berlin, Heidelberg, 2009: pp. 79–96. doi: 10.1007/978-3-540-87395-2_6.
- [62] OWL Working Group, PrimerExampleTurtle - OWL, (2008). <https://www.w3.org/2007/OWL/wiki/PrimerExampleTurtle> (accessed May 18, 2023).
- [63] RDFLib Team, rdflib 6.2.0, (2022). <https://rdflib.readthedocs.io/> (accessed May 18, 2023).
- [64] Santosh Philip, Welcome to eppy's documentation!, (2022). <https://eppy.readthedocs.io/> (accessed May 18, 2023).
- [65] B. Cho, T. Dayrit, Y. Gao, Z. Wang, T. Hong, A. Sim, K. Wu, Effective missing value imputation methods for building monitoring data, in: 2020 IEEE International Conference on Big Data (Big Data), IEEE, 2020: pp. 2866–2875.
- [66] Big Ladder Software LLC, EnergyPlus Weather File (EPW) Data Dictionary, (2023). <https://bigladdersoftware.com/epx/docs/8-2/auxiliary-programs/energyplus-weather-file-epw-data-dictionary.html> (accessed May 18, 2023).
- [67] Stanford Center for Biomedical Informatics Research, Protégé is a free, open-source ontology editor and framework for building intelligent systems, (2020). <https://protege.stanford.edu/> (accessed May 18, 2023).
- [68] U.S. Department of Energy's (DOE) Building Technologies Office (BTO), National Renewable Energy Laboratory (NREL), EnergyPlus: Weather data, (2023). <https://energyplus.net/weather> (accessed May 18, 2023).
- [69] Hong Kong Observatory, The Government of the Hong Kong Special Administrative Region, Hong Kong Observatory, (2021). <https://www.hko.gov.hk/en/index.html> (accessed May 18, 2023).
- [70] Autodesk Inc, Autodesk Revit Features, (2023). <https://www.autodesk.com/products/revit/features> (accessed May 18, 2023).
- [71] Delta Controls Inc., Delta Controls, (2023). <https://deltacontrols.com/> (accessed May 18, 2023).
- [72] U.S. Department of Energy's (DOE) Building Technologies Office (BTO), National Renewable Energy Laboratory (NREL), EnergyPlus: Downloads, (2023). <https://energyplus.net/downloads> (accessed May 18, 2023).
- [73] M.K. Bracht, A.P. Melo, R. Lamberts, A metamodel for building information modeling-building energy modeling integration in early design stage, Autom. Constr. 121 (2021), 103422, <https://doi.org/10.1016/j.autcon.2020.103422>.
- [74] S. Ali, Analysis of Procedures and Workflow for Conducting Energy Analysis Using Autodesk Revit, gbXML, and Trace700, in: Proceedings of SimBuild Conference 2010: 4th Conference of IBPSA-USA, IBPSA-USA, New York City, USA, 2010: pp. 56–63. https://publications.ibpsa.org/conference/paper/?id=simbuild2010_TS02A-02-Ali.
- [75] H. Kim, K. Anderson, Energy modeling system using building information modeling open standards, J. Comput. Civ. Eng. 27 (2013) 203–211, [https://doi.org/10.1061/\(ASCE\)CP.1943-5487.0000215](https://doi.org/10.1061/(ASCE)CP.1943-5487.0000215).
- [76] D.A.L. Calquin, G. Wandersleben, L.S. Castillo, Interoperability map between BIM and BPS software, Comput. Civ. Build. Eng. 2014 (2014) 601–608, <https://doi.org/10.1061/9780784413616.075>.
- [77] V. Dimitriou, S.K. Firth, T.M. Hassan, F. Fouchal, BIM enabled building energy modelling: development and verification of a GBXML to IDF conversion method, in: In Proceedings of the 3rd IBPSA-England Conference BSO 2016, Great North Museum, Newcastle, UK, 2016. <https://www.semanticscholar.org/paper/BIM-enabled-building-energy-modelling%3A-development-Dimitriou-Firth/93717bfb6dff5dc2e64dc928453874c4dfc067a0>.